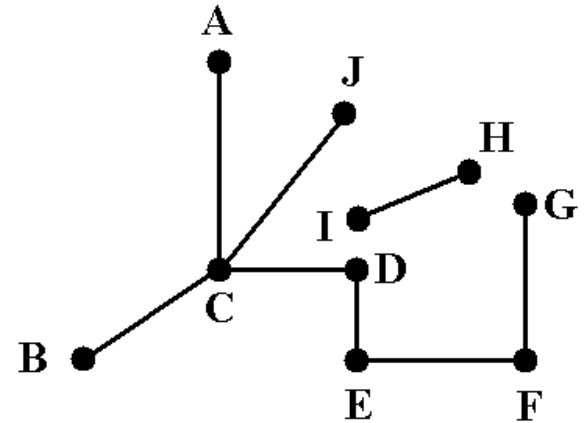


Discrete dynamic modeling of signaling networks

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The signaling networks considered here are not known at the level of elementary reactions, thus they are poor candidates for biochemical modeling approaches.

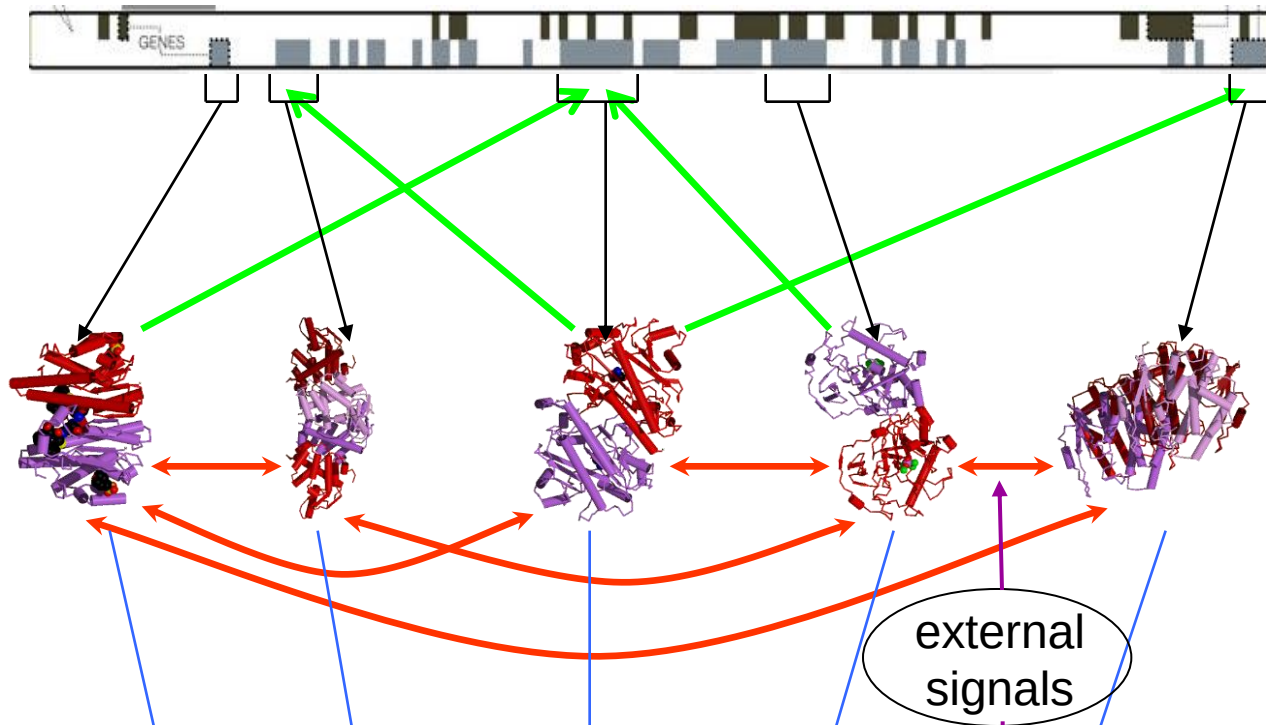


Network (graph): a set of nodes (vertices) connected pairwise by edges (links).

- The nodes have specific identities, e.g. a molecular species
- Each node has an abundance (state) associated with it
- The abundances can change in time or by context

Need extensions of graph theory: different types of nodes and edges, cooperative (hyper) edges, node/edge weights, edge signs.

Many **non-identical** elements, **diverse** interactions



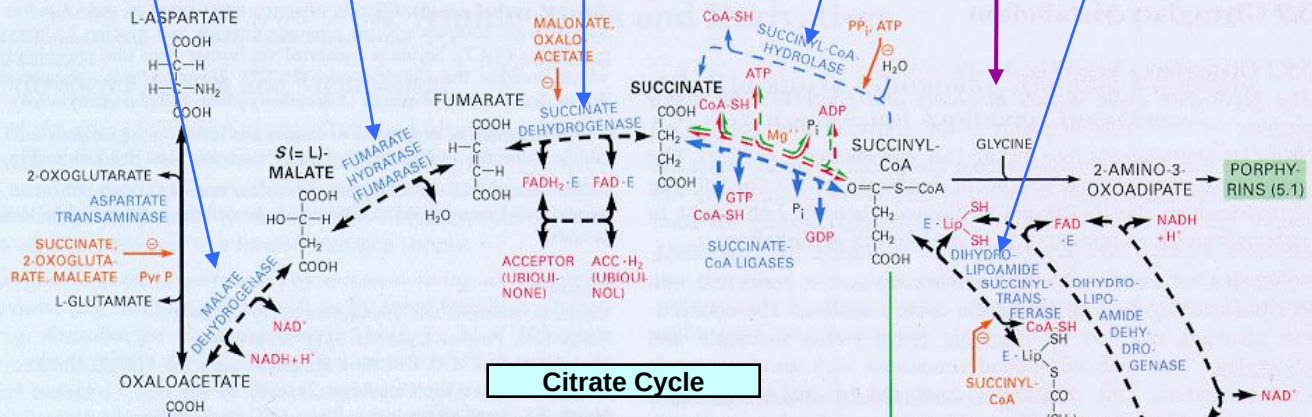
GENOME
gene regulation

PROTEOME
protein-protein
interactions

signal transduction

METABOLISM

Bio-chemical
reactions

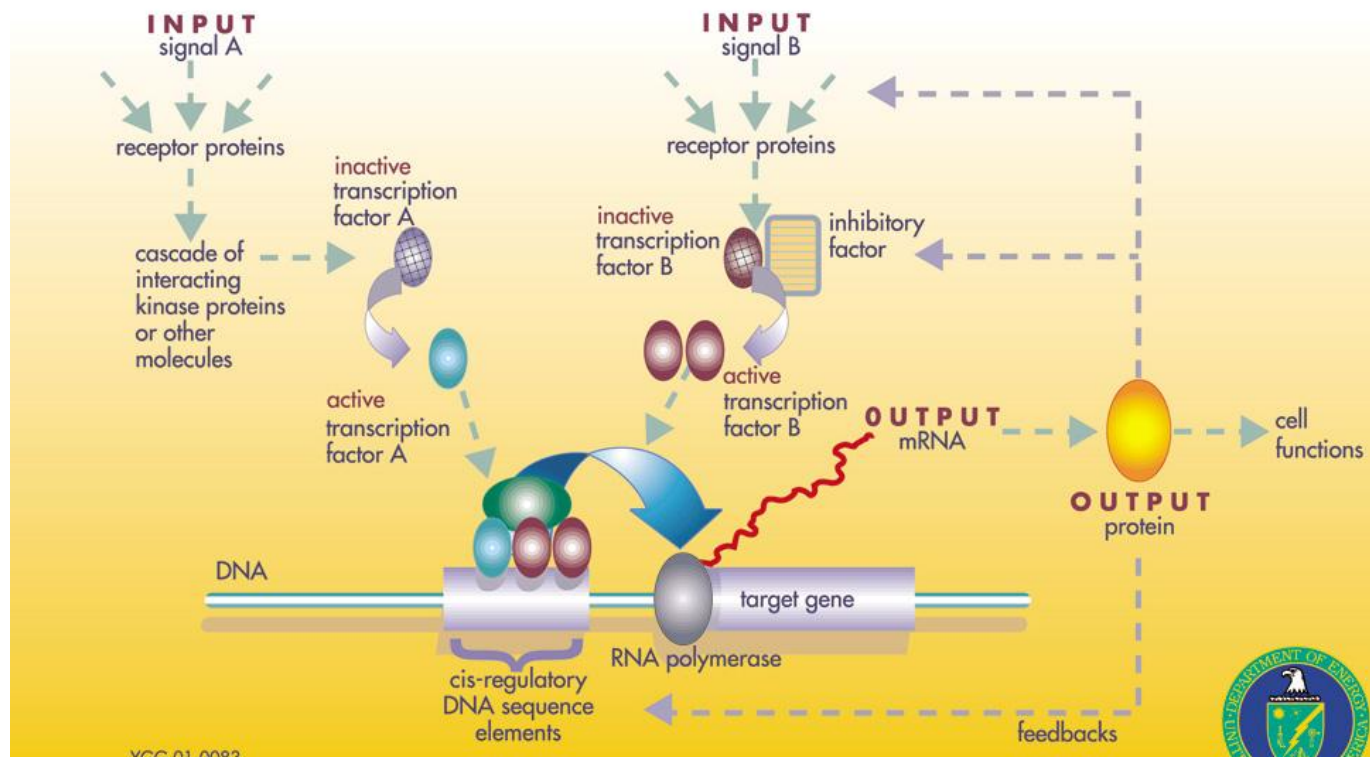


Signaling networks

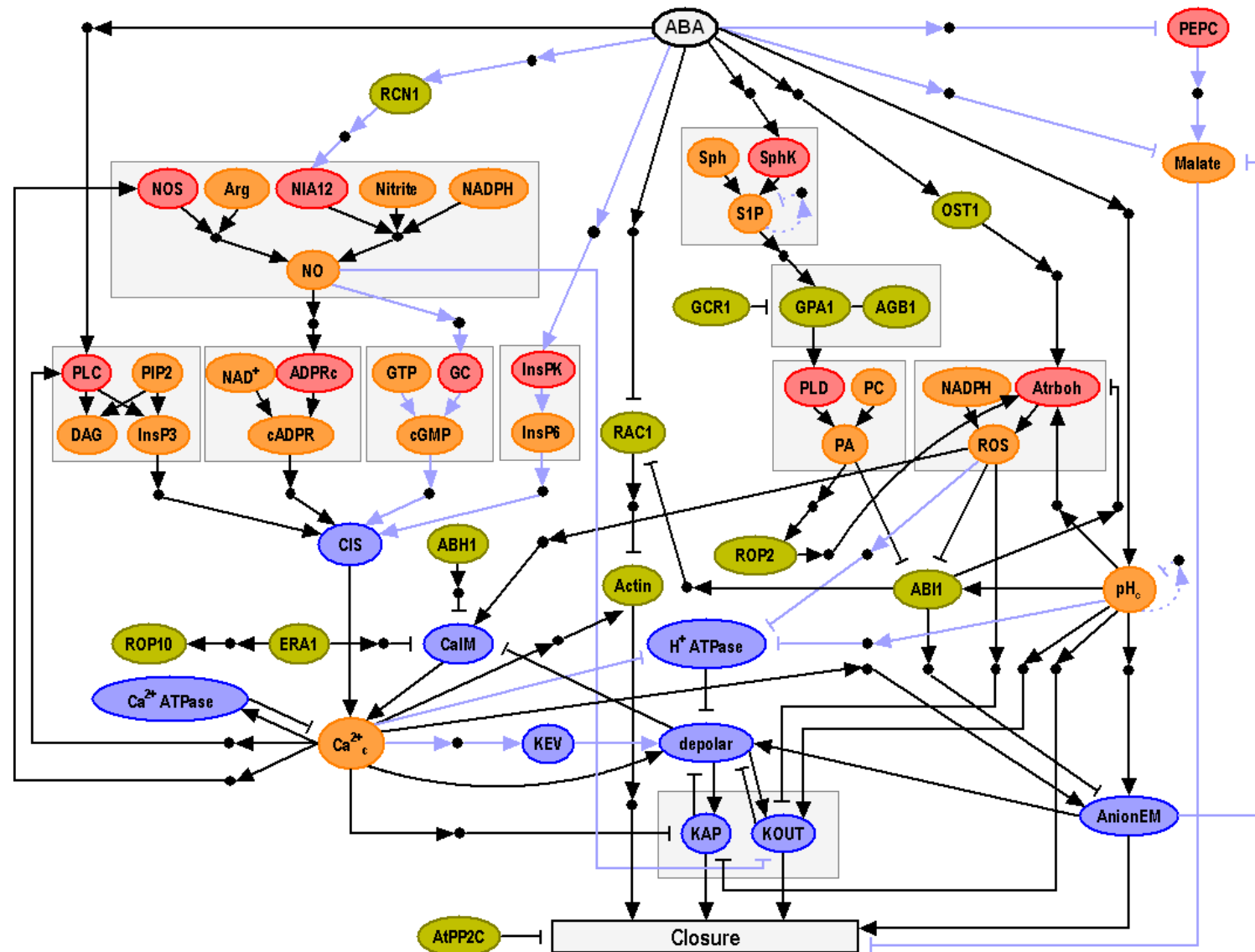
Nodes: proteins, molecules, cells

Edges: reactions and processes (e.g. ligand/receptor binding, protein conformational changes, ligand production); common to all is that they reflect **information transfer**

Signal transduction networks have defined inputs and outputs.



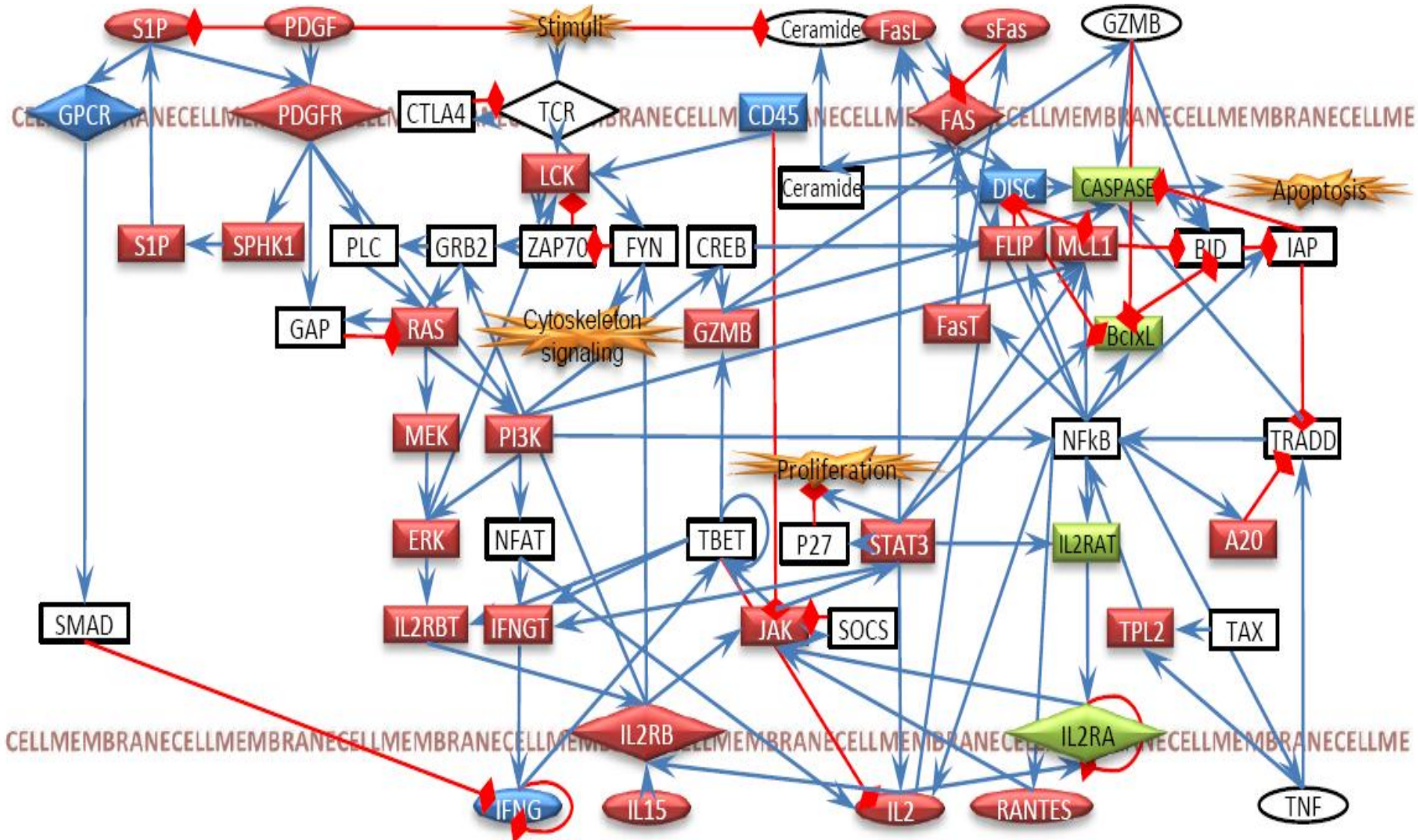
ABA signal transduction network



Red: enzymes
 Blue: transport
 Orange: small molecules
 Green: sign. transd. proteins
 Black points: unknown intermediary nodes

Li, Assmann, Albert, PLoS Biology 2006

T-LGL survival signaling network



Rectangle: intracellular; ellipse: extracellular; diamond: receptor.

Upregulated, downregulated, deregulated node; Activation, inhibition edge

Modeling the dynamics of biological systems

First: define the system; collect known states or behavior

Input: components; states of components; interactions

Hypotheses: interaction network, interaction kinetics.

Validation: capture known dynamic behavior.

Explore: study cases that are not accessible experimentally

- change parameters

- change assumptions

Types of models: continuous or discrete
 deterministic or stochastic

Continuous and deterministic models - > medium node abundances,
< medium-size networks.

Stochastic models - low node abundances, small networks

Discrete models – multi-modal node abundances, > medium networks.

Discrete dynamic modeling of biological systems

- The functional form of regulatory relationships and the kinetic parameters are often unknown
- Increasing evidence for
 - robustness to changes in kinetic parameters
 - sigmoidal regulatory functions
 - bistability (two steady states)

Hypothesis: the kinetic details of individual interactions are less important than the organization of the regulatory network

Discrete dynamic models assume that nodes can be characterized by only a few (minimum two) discrete states.

The states are governed by the interactions encapsulated in the network.

Boolean modeling of biological systems

Main assumption: components have two main states :

Expressed or not expressed, active or inactive, open or closed (ion channel), high or low level. Denote these states by ON (1) or OFF (0)

The future state of a regulated node (the output) depends on the current state of its regulators (inputs), which may or may not include its own current state.

e.g. If transcription factor is active, gene will be transcribed,
gene will be expressed after a certain time.

Node-specific transfer functions specify the input-output relationships

- Boole logic: based on the operators NOT, AND, OR
- Threshold model: output is given by comparing the weighted sum of inputs to a threshold

Examples of transfer functions

<i>In</i>	<i>Out</i>
0	1
1	0

Out = NOT *In*

<i>In1</i>	<i>In2</i>	<i>Out</i>
0	0	0
0	1	0
1	0	0
1	1	1

Out = *In1* AND *In2*

<i>In1</i>	<i>In2</i>	<i>Out</i>
0	0	0
0	1	1
1	0	1
1	1	1

Out = *In1* OR *In2*

$$Out = \begin{cases} 1, & \text{if } -In \geq 0 \\ 0, & \text{if } -In < 0 \end{cases}$$

$$Out = \begin{cases} 1, & \text{if } In1 + In2 \geq 2 \\ 0, & \text{if } In1 + In2 < 2 \end{cases}$$

$$Out = \begin{cases} 1, & \text{if } In1 + In2 \geq 1 \\ 0, & \text{if } In1 + In2 < 1 \end{cases}$$

From dose-response curves to Boolean switches

- X – mRNA
- Y – transcriptional activator

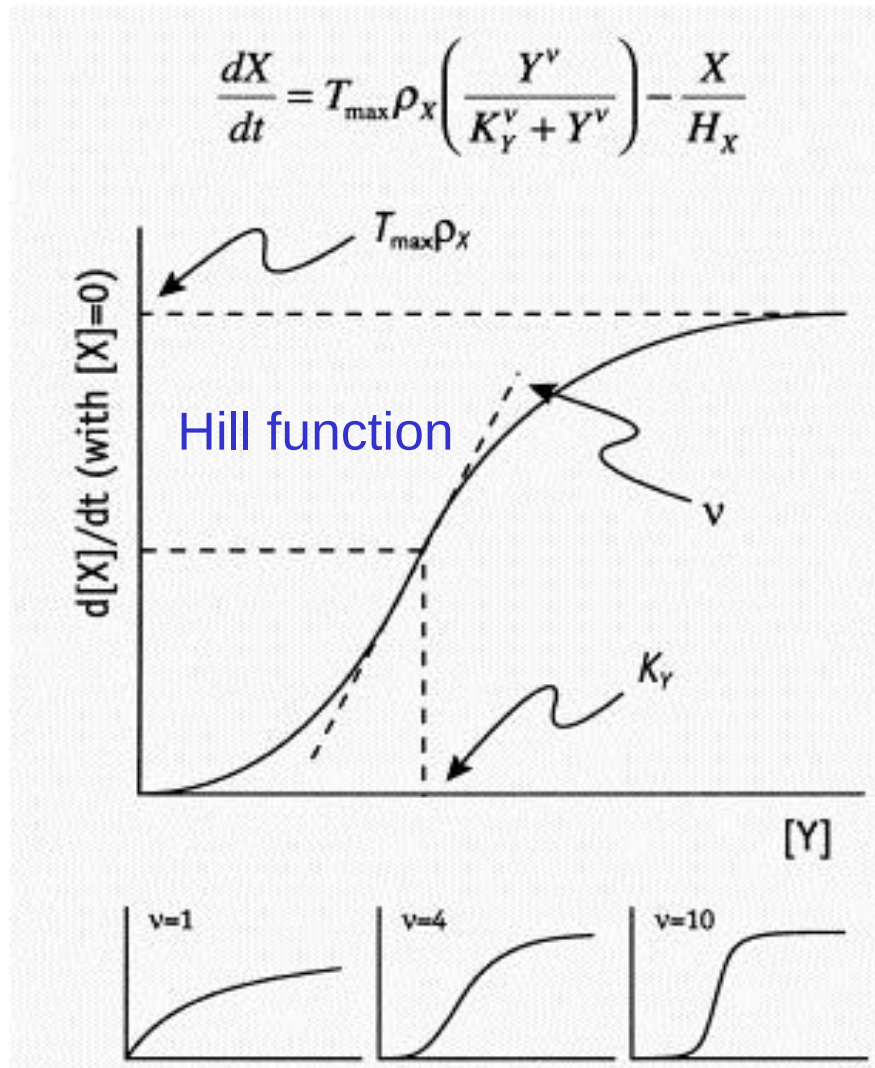
If v is large, the synthesis function approaches

$$T_{\max}\rho_X B(Y)$$

$B(Y)=1$ if $Y \geq K_Y$ and 0 otherwise

If $Y \geq K_Y$ $dX/dt \approx T_{\max}\rho_X - X/H_X$
Asymptotic increase toward $T_{\max}\rho_X H_X$

If $Y < K_Y$ $dX/dt \approx -X/H_X$
Exponential decrease to zero.



Hybrid models: Boolean regulation combined with continuous decay

- Each node is characterized by both a continuous variable (concentration) and a Boolean variable (activity).

$$\frac{d\hat{X}_i}{dt} = B_i(X_1, X_2, \dots) - \hat{X}_i$$

- X_i is defined by the threshold rule

$$X_i = \begin{cases} 0, & \text{if } \hat{X}_i < 0.5 \\ 1, & \text{if } \hat{X}_i \geq 0.5 \end{cases}$$

- Compared to $\frac{dX}{dt} = T_{\max} \rho_X \left(\frac{Y^v}{K_Y^v + Y^v} \right) - \frac{X}{H_X}$ this assumes

activation threshold=0.5, maximal synthesis rate = decay rate= 1

Hybrid model of a positive feedback loop

$$\frac{d\hat{A}}{dt} = C - \hat{A} \quad \frac{d\hat{B}}{dt} = A - \hat{B} \quad \frac{d\hat{C}}{dt} = B - \hat{C} \quad A, B, C = \begin{cases} 0, & \text{if } \hat{A}, \hat{B}, \hat{C} < 0.5 \\ 1, & \text{if } \hat{A}, \hat{B}, \hat{C} \geq 0.5 \end{cases}$$

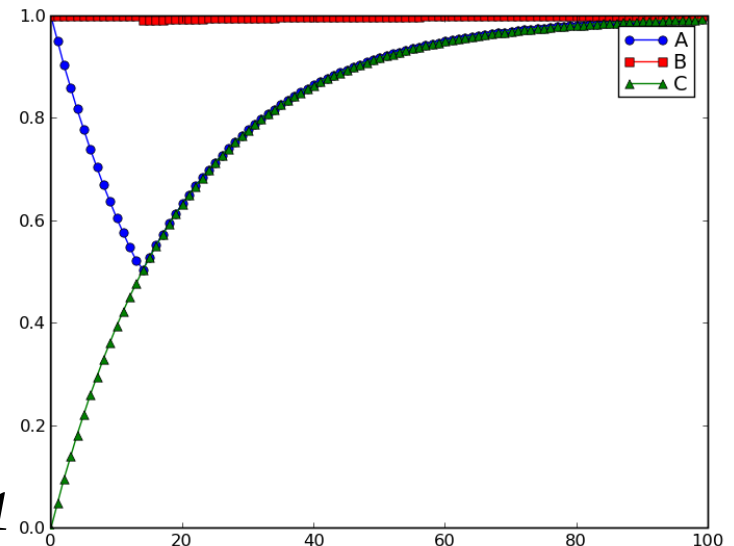
Initial condition: $A = B = \hat{A} = \hat{B} = 1 \quad C = \hat{C} = 0$

$$\begin{aligned} \frac{d\hat{A}}{dt} &= -\hat{A} & \hat{A} &= e^{-t} \\ \frac{d\hat{B}}{dt} &= 1 - \hat{B} & \hat{B} &= 1 \\ \frac{d\hat{C}}{dt} &= 1 - \hat{C} & \hat{C} &= 1 - e^{-t} \end{aligned}$$

until $t_1 = \ln 2$ when $C=1$

$$\frac{d\hat{A}}{dt} = 1 - \hat{A} \quad \hat{A} = 1 - e^{t_1 - t}$$

Steady state: $A = B = C = \hat{A} = \hat{B} = \hat{C} = 1$



Hybrid model of a negative feedback loop

$$\frac{d\hat{A}}{dt} = \text{not } B - \hat{A} \quad \frac{d\hat{B}}{dt} = A - \hat{B} \quad A, B = \begin{cases} 0, & \text{if } \hat{A}, \hat{B} < 0.5 \\ 1, & \text{if } \hat{A}, \hat{B} \geq 0.5 \end{cases}$$

Initial condition: $A = B = \hat{A} = \hat{B} = 0$

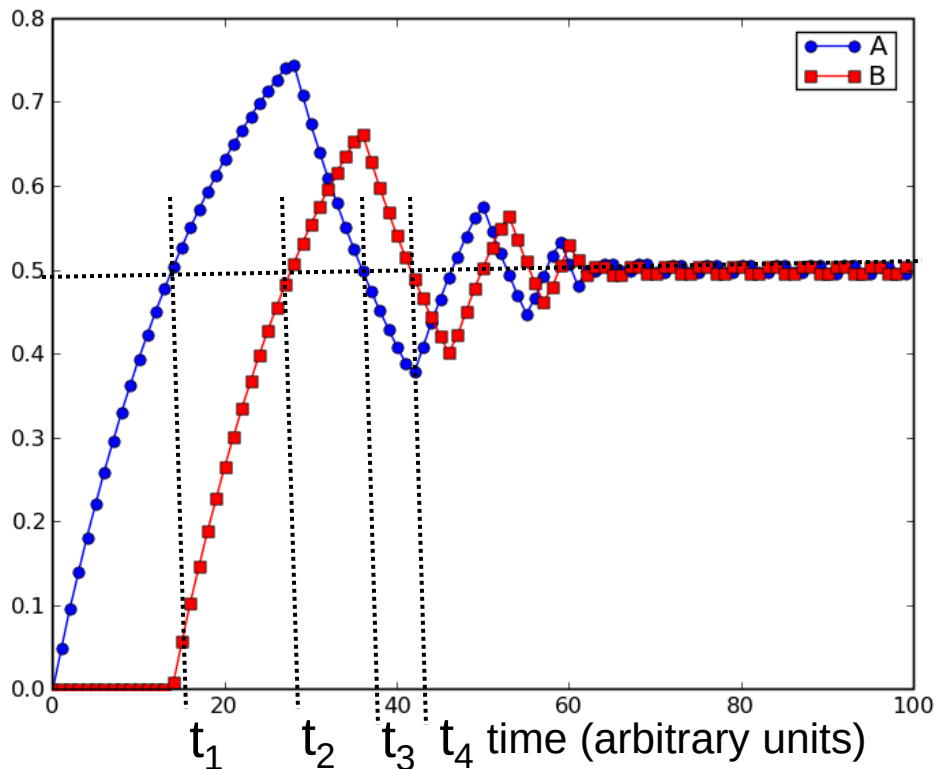
$$\begin{aligned} \frac{d\hat{A}}{dt} &= 1 - \hat{A} & \hat{A} &= 1 - e^{-t} & \text{until } t_1 = \ln 2 & \text{when } A=1 \\ \frac{d\hat{B}}{dt} &= -\hat{B} & \hat{B} &= 0 \end{aligned}$$

$$\frac{d\hat{B}}{dt} = 1 - \hat{B} \quad \hat{B} = 1 - e^{t_1 - t} \quad \text{until } t_2 = 2 \ln 2 \quad \text{when } B=1$$

$$\frac{d\hat{A}}{dt} = -\hat{A} \quad \hat{A} = 0.75e^{t_2 - t} \quad \text{until } t_3 = \ln 2 + \ln 3 \quad \text{when } A=0$$

Simulation result

Used the software library BooleanNet <http://code.google.com/p/booleannet/>



$0 \leq t < t_1: A=B=0$

$t_1 \leq t < t_2: A=1, B=0$

$t_2 \leq t < t_3: A=B=1$

$t_3 \leq t < t_4: A=0, B=1$

$t_4 \leq t < t_5: A=B=0$

.....

Discrete state transitions at discrete time points.

It may be sufficient to characterize the state transitions without explicitly considering time.

From dose-response curves to Boolean switches

- X – mRNA
- Y – transcriptional activator

Boolean simplification:

X and Y can only have two states: ON (above threshold) and OFF (below threshold)

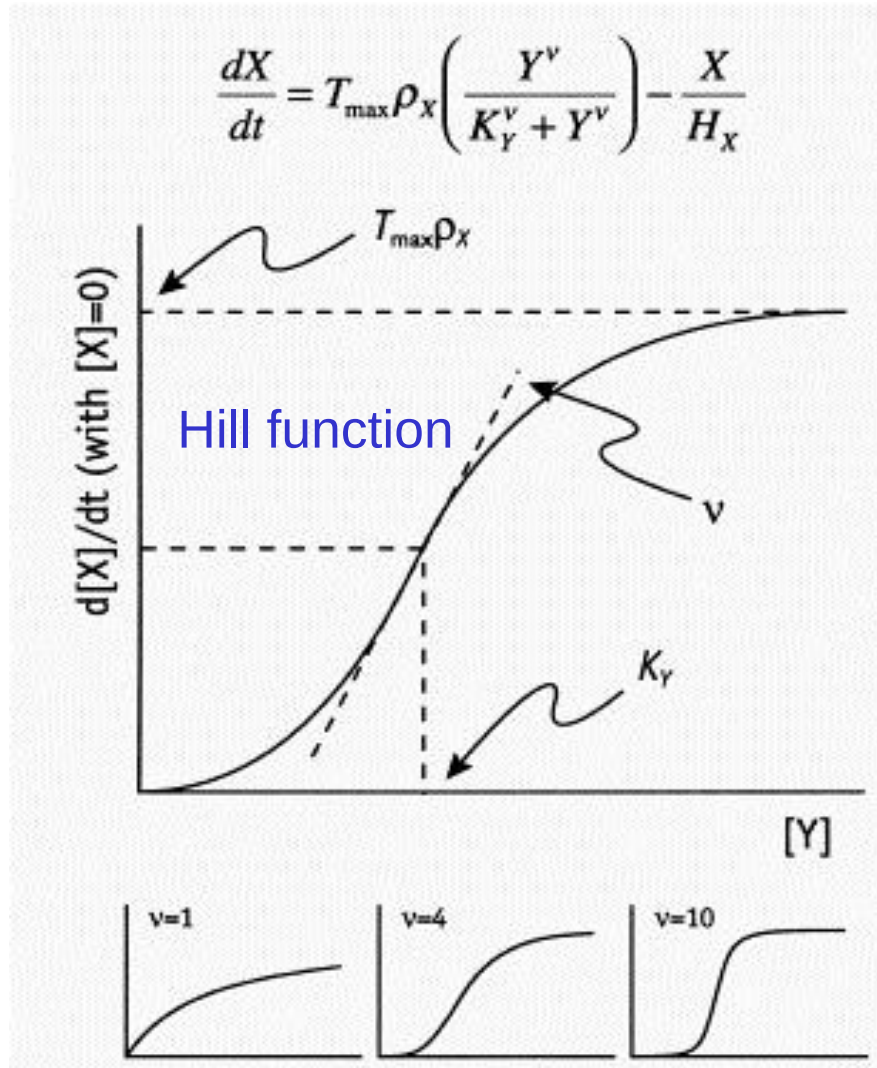
$$X^* = Y$$

Activation:

If Y=ON $X^*=$ ON

Decay:

If Y= OFF $X^*=$ OFF



Implementing time in discrete models

1. Synchronous models

- The state of each node is updated simultaneously at multiples of a common timestep.
- Thus the future state means the state at the next timestep
- Underlying assumption: the timescales of all synthesis and decay processes are similar

2. Asynchronous models

The state of each node is updated individually

Implementations:

- Different update time for each node $T_i^k = k\gamma_i$
- Select a random update order (permutation) in each timestep.
- Update a randomly selected node in each timestep.

Ex.1

Let's consider a three-node network. Implement the three asynchronous update methods to give a sequence of 10 node updates.

Which update method is deterministic and which is stochastic?

Ex. 2

Let's formulate the Boolean correspondent of the negative feedback loop hybrid model example.

What type of update will give the same state sequence as the hybrid model?

Steps in constructing a Boolean model of a regulatory or signaling network

1. Input to the model: components; interactions; binary states of components in certain known conditions
2. Construct the interaction network
3. Determine the Boolean transfer functions for each node
4. Determine the relevant initial conditions
5. Analyze the dynamics of the model
 - determine the attractors and their basins of attraction
 - determine the effects of node perturbations
6. Compare with known behavior. If there are discrepancies, revise the network and/or the Boolean transfer functions
7. Use the model to make novel predictions.

Ex. 3

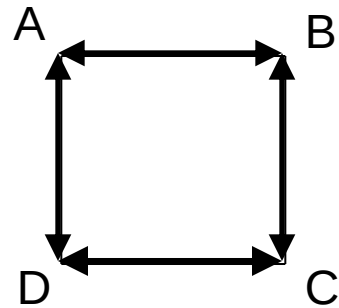
Construct a network of three nodes, such that each node has at least one incoming edge. Associate a Boolean rule to each node. Assume that each node's state is updated at the same time (synchronous update). Start with an initial state, and update the state of the nodes 10 times.

What is happening to the state of the network?

Start from a different initial state. Is the final behavior be the same?

How many different final states/behaviors can the network have?

Example

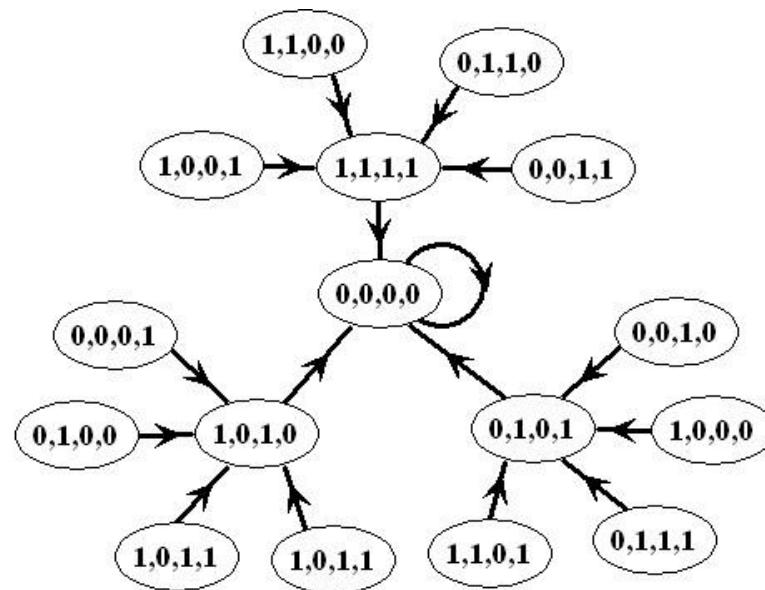


I1	I2	O
0	0	0
0	1	1
1	0	1
1	1	0

States

A	B	C	D
0	0	0	0
0	0	0	1
....	...	...	...
1	1	1	1

Transitions



Concepts in Boolean network dynamics

Attractor – a set of states that repeats itself, can be a **fixed point** or a **complex attractor**

Fixed point: Future State = Current State = Previous State

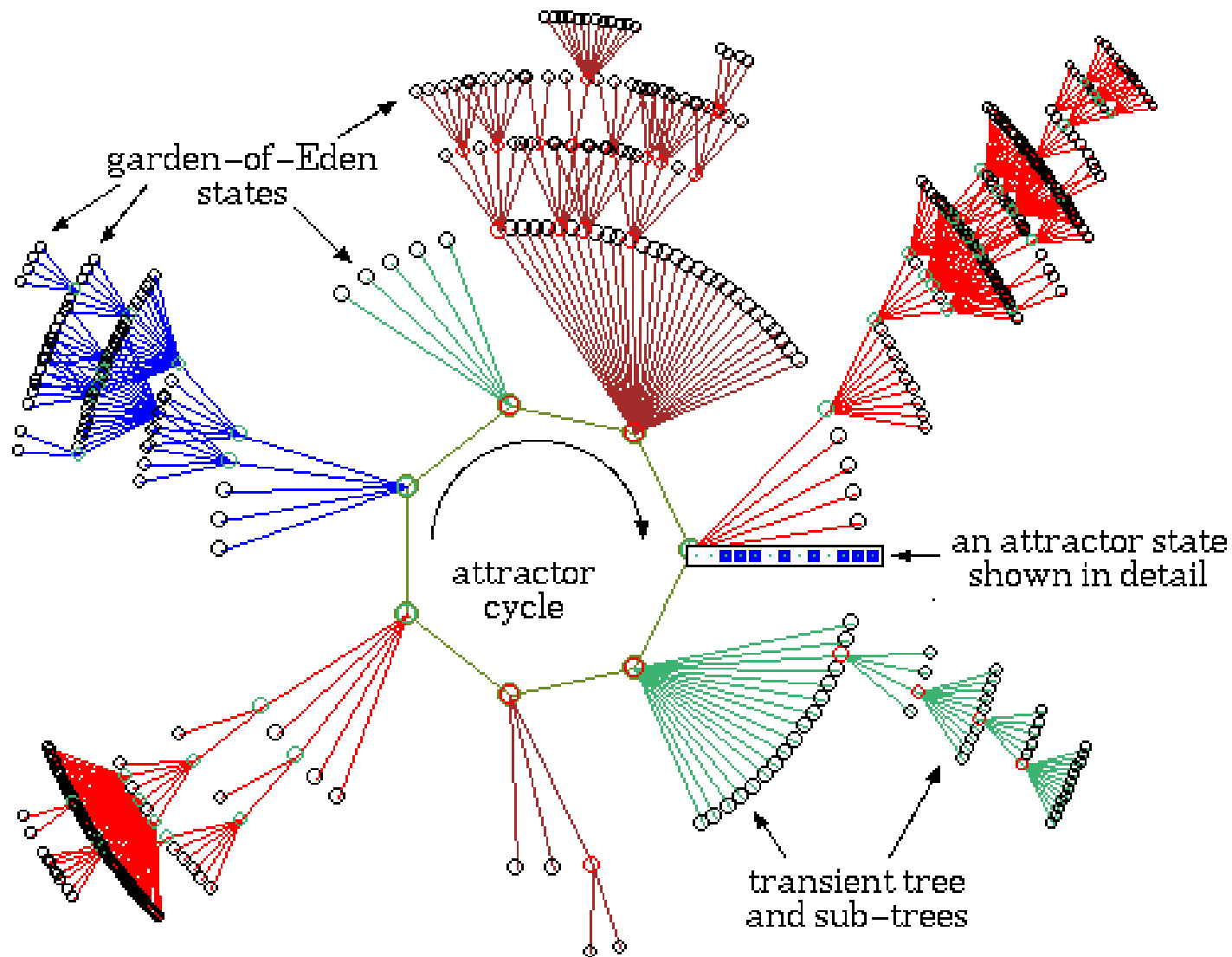
All states lead to or are part of an attractor.

Basin of attraction – all states leading to a given attractor

In a network of N nodes the maximum possible size of an attractor is the total number of states, 2^N

In practice, the size of the attractor is much shorter than this maximum.

State transition graph for a synchronous Boolean model



Ex. 4. Consider a network of four nodes. Node A is the signal, the Boolean rules of the other three nodes are the following:

$$B^* = A \text{ or } C, \quad C^* = A \text{ and not } D, \quad D^* = B \text{ and } C$$

Set $A = 0$.

(a) Assume that each node's state is updated at the same time (synchronous update). Start with an initial state, and update the state of the nodes 5 times.

What attractor did you find?

(b) Now start from the same initial state, but update only one, randomly selected, node at each step.

Is the result the same?

Q. How can you determine the fixed points of a Boolean model without performing updates?

In the fixed point time does not matter, thus the transfer functions become equations

$$B = A \text{ or } C$$

$$C = A \text{ and not } D$$

$$D = B \text{ and } C$$

$$\text{Solution: } A = 0, B = 0, C = 0, D = 0$$

Asynchronous models have the same fixed points as synchronous ones.

Ex. 5. Consider a the same network of four nodes.

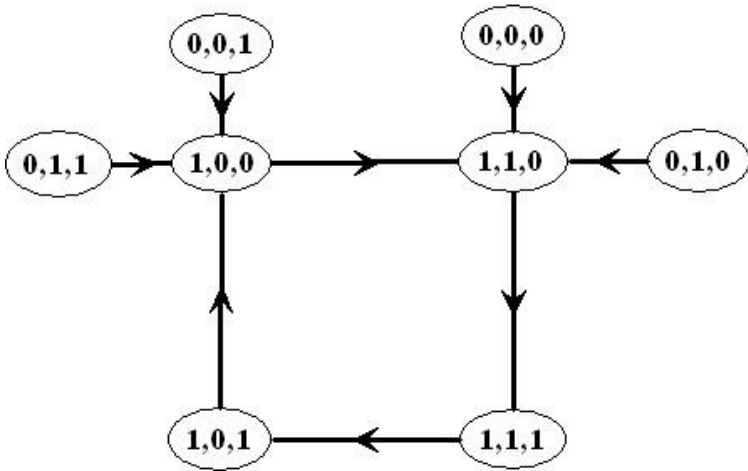
$B^* = A \text{ or } C$, $C^* = A \text{ and not } D$, $D^* = B \text{ and } C$

Set $A = 1$.

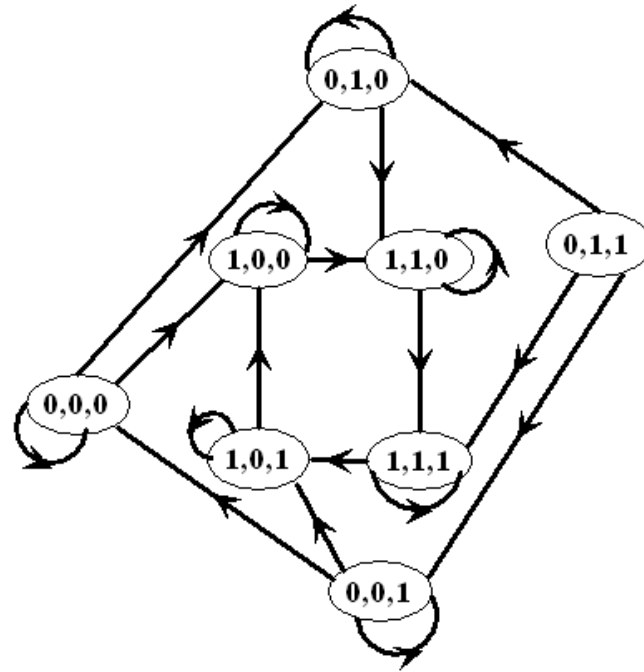
- (a) Assume that each node's state is updated at the same time (synchronous update). Start with an initial state, and update the state of the nodes 10 times. What attractor did you find?
- (b) Now start from the same initial state, but update a randomly selected node at each step. Is the attractor the same?
- (c) Construct the state transition graph of the synchronous model.

What changes do you expect when switching to asynchronous update?

Attractors for synchronous and asynchronous models



Synchronous



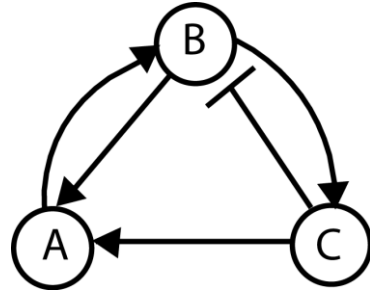
General asynchronous

The analog of a periodic orbit in a synchronous model is a strongly connected component that has an empty out-component in an asynchronous model.

I. Harvey, T. Bossomaier, Proc. ECAL97, 67 1997

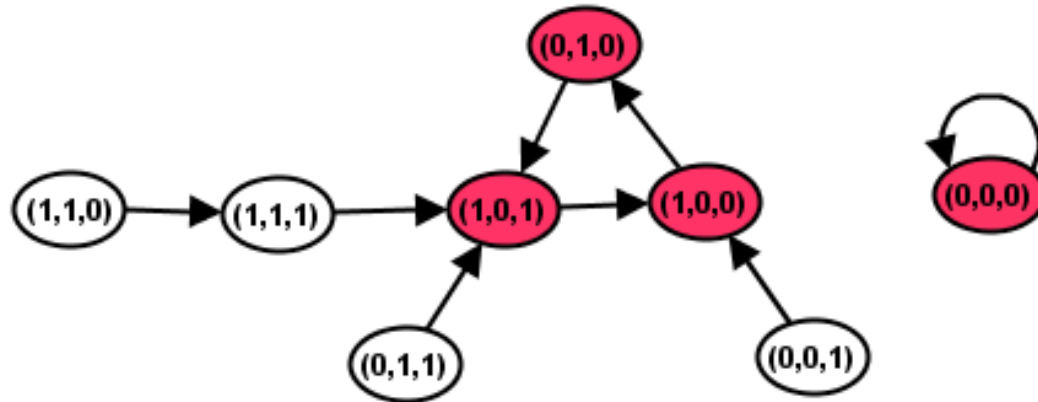
Comparison of state transition graphs of synchronous and asynchronous models

Example



Node	Boolean rule
A	$A^* = B \text{ OR } C$
B	$B^* = A \text{ AND } (\text{NOT } C)$
C	$C^* = B$

Synchronous state transition graph:

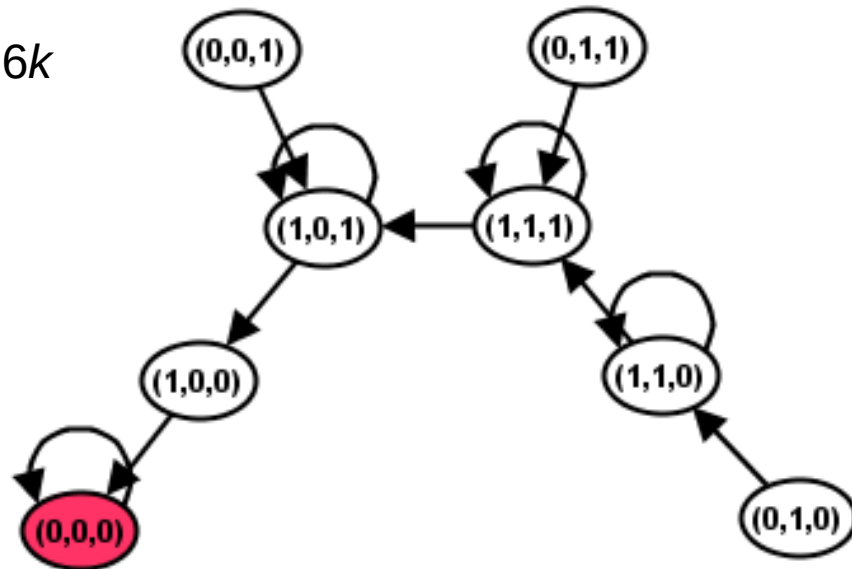


- Each state has one and only one possible successor
- There is a fixed point and an extended attractor (limit cycle) of size 3

Deterministic asynchronous state transition graph

Each node has an associated time unit and is updated at multiples of its time unit

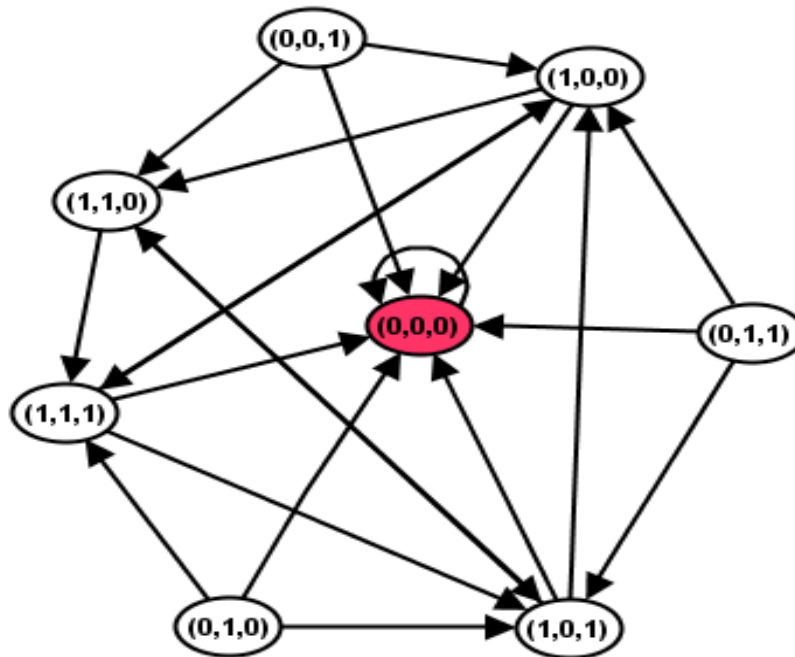
Given $T_A = 2k$, $T_B = 3k$ and $T_C = 6k$



- Self loops appear when a node is updated but its state does not change
- For the given time units, only the fixed point of synchronous model is reached. There might exist time units that lead to complex attractors.

Random order asynchronous state transition graph

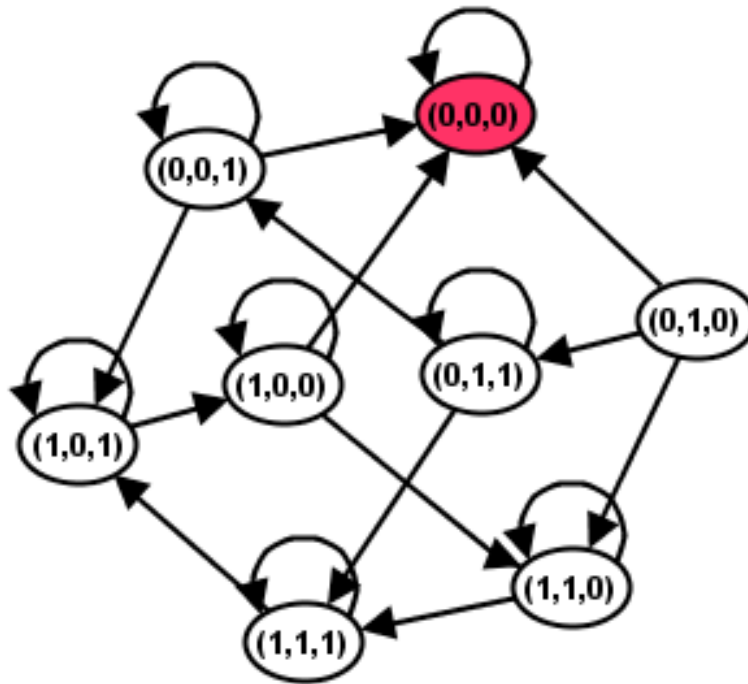
Select a random update order in each time step and update the nodes in that order



- Each state can have multiple successors
- In this particular example, only the fixed point of the synchronous model is reached

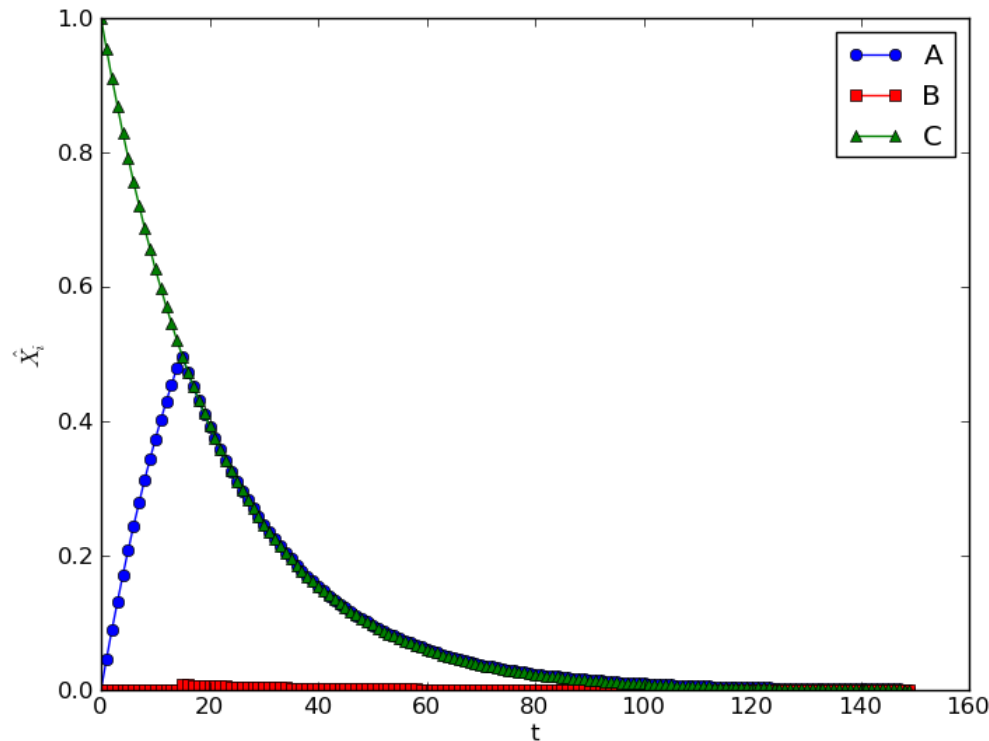
General asynchronous state transition graph

Update a randomly selected node in each time step



- Each state can have multiple successors
- Self loops appear when a node is updated but its state does not change
- In this particular example, only the fixed point of synchronous model is reached

Hybrid model for initial state $A=B=0$ and $C=1$



- The fixed points of the hybrid model are the same as those of the corresponding Boolean model

Synchronous, asynchronous and hybrid models can be implemented in BooleanNet <http://code.google.com/p/booleannet/>

Example 1: Modeling drought signaling in plants

Phenomenon: abscisic acid induced closure of plant stomata

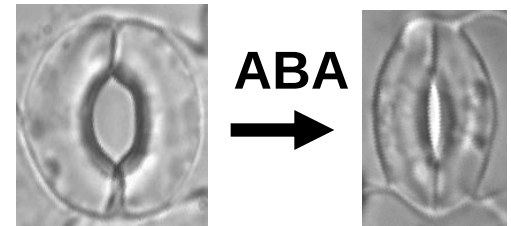
Hypotheses: network inference from indirect information
protein activity is switch-like

Validation: reproduces known wild type and disrupted behavior.

Explored: disruptions

changes in initial conditions

changes in timing



Insight: variability in timing and initial conditions does not matter
65% of perturbations have no negative effect
identified critical perturbations

S. Li, S. Assmann and R. Albert, PLoS Biology 4, e312 (2006).

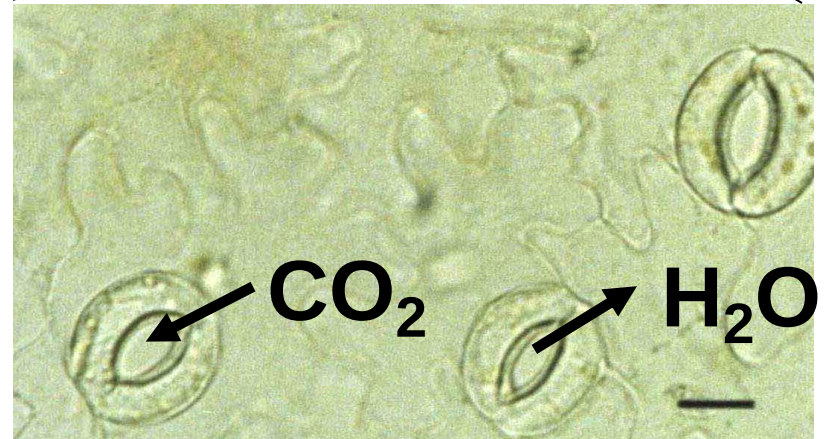
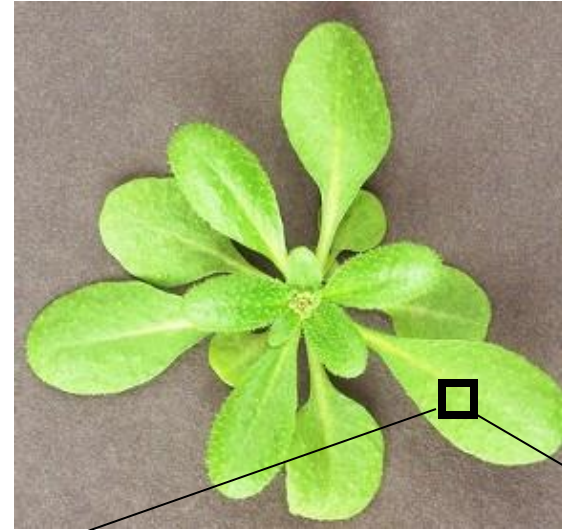
Stomata and guard cells

The exchange of O_2 and CO_2 in plants occurs through **stomata**.

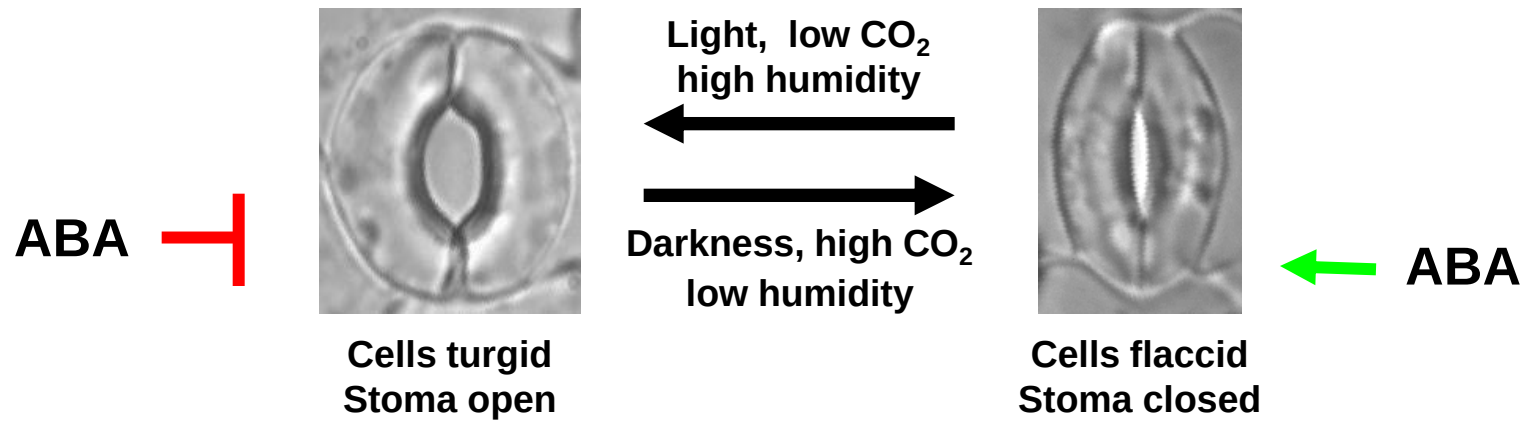
90% of the water taken up by a plant is lost in transpiration.

Stomatal sizes are determined by the turgor (fullness) of the **guard cells**.

The latter is orchestrated by a complex **signal transduction network**.



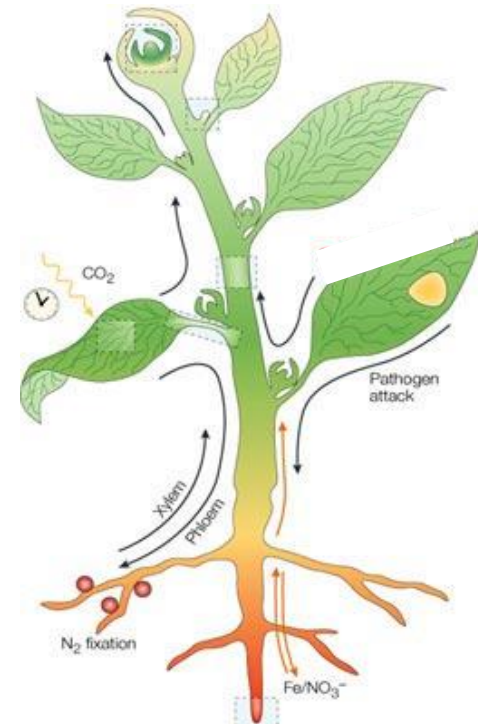
Modeling abscisic acid (ABA) signaling in plants



During drought conditions plants synthesize a hormone called abscisic acid (ABA).

ABA initiates a signal transduction network of its own to **close stomata**.

More than 20 proteins, molecules, channels implicated, but most evidence is indirect.



Network construction from indirect evidence

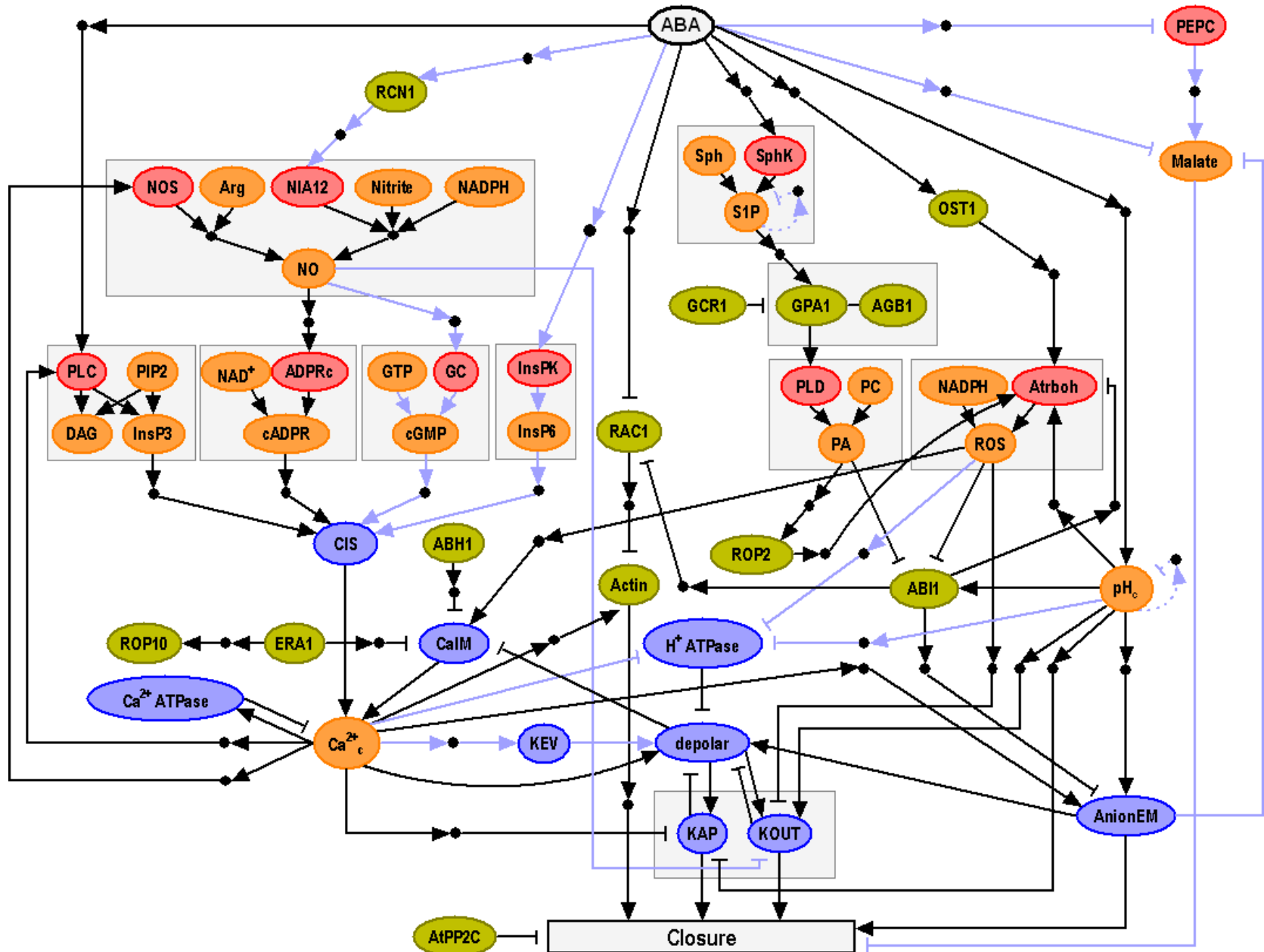
Nodes: all proteins, chemicals, ion channels implicated in the process
Compressed biological information into activation or inhibition

Hypotheses: indirect causal relationships correspond to **paths**;
activating or inhibiting effects on processes represented as two
paths that **intersect** at an unknown intermediary node

Found the most parsimonious (least redundant) network that incorporates all nodes and known processes.

Algorithm later formalized and implemented in NET-SYNTHESIS
<http://www.cs.uic.edu/~dasgupta/network-synthesis/>

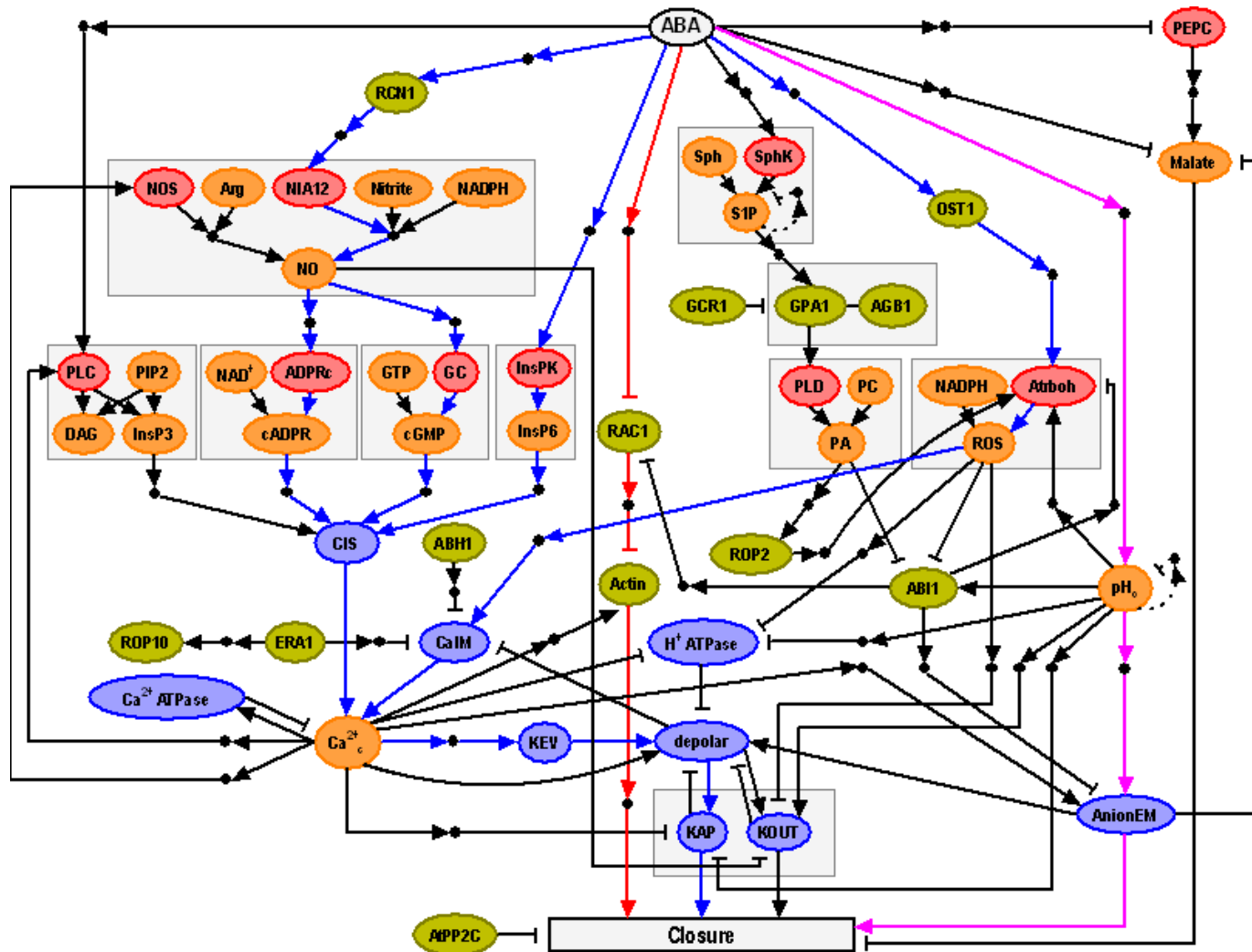
R. Albert, B. DasGupta et al, Journ. Comp Biology 14, 927 (2007).
S. Kachalo et al., Bioinformatics 24, 293 (2008).



enzymes, signal transduction proteins, transport, small molecules

• intermediary nodes, — inferred edges

Topologically redundant paths are in fact conditional



At least three separate ABA-closure paths, through Ca^{2+} , through pH_c and through actin

Collected evidence of conditionality from literature

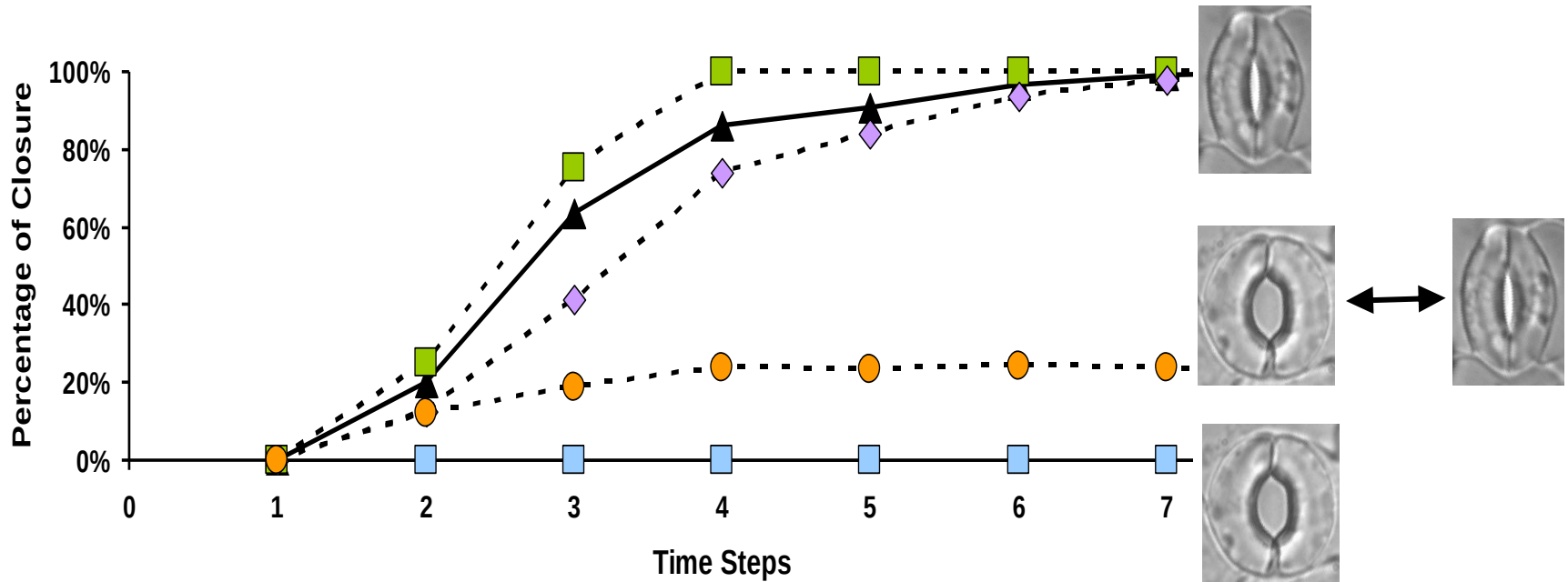
$\text{Ca}^{2+}_c^* = (\text{CaIM or CIS})$ and not $\text{Ca}^{2+} \text{ATPase}$

$\text{Closure}^* = (\text{KOUT or KAP})$ and AnionEM and Actin and not Malate

Handling the lack of timing and state information

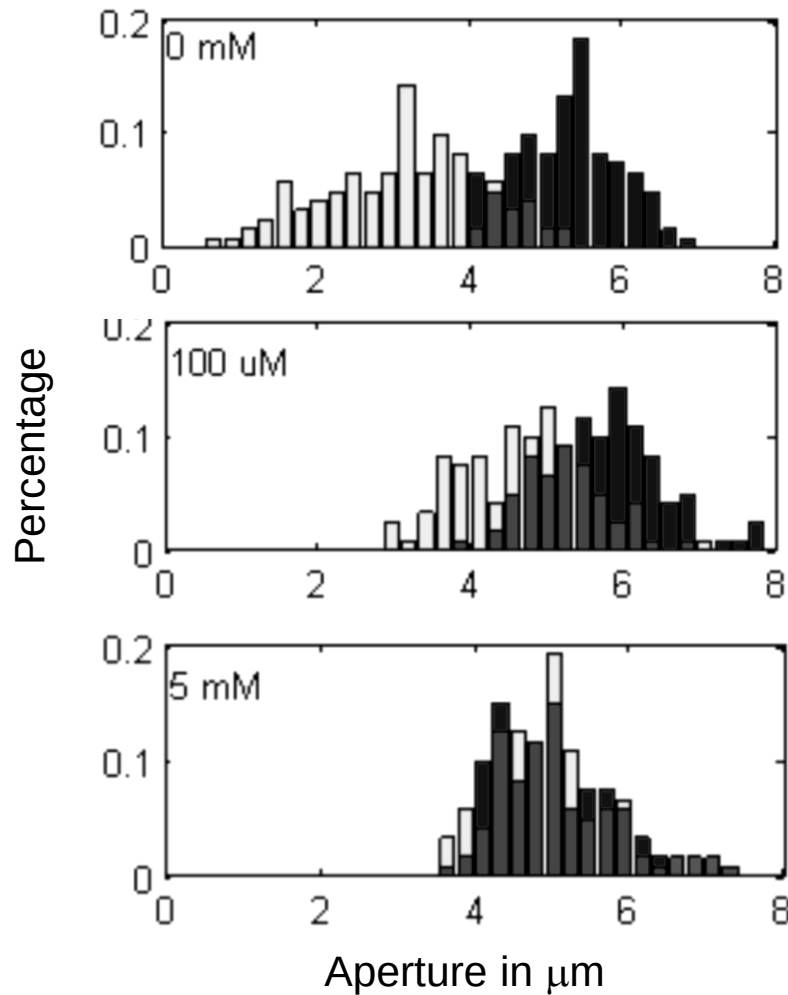
- Set internal nodes randomly to 0 or 1 to mimic the noise in the internal environment of the guard cell.
- Apply ABA=1 .
 - In a random order, update every node according to its Boolean rules.
 - Check the status of the Closure node (1st timestep)
 - In a different random order, update every node according to its Boolean rules.
 - Check the status of the Closure node (2nd timestep).
 - Etc.
- Do this for 10,000 randomly chosen starting conditions.
- At each timestep, calculate the percentage of Closure=1 in this population of 10,000 in silico guard cells = **Percentage of Closure**.
- Carry out the same simulation with individual nodes disrupted (set to 0) to mimic knockout mutations, etc.

Results: dynamic effects of knockouts



- ▲ Normal response of simulated population to ABA stimulus.
- ABI1 knockout mutants respond faster.
- ◆ Perturbations in Ca^{2+} lead to slower response at the population level.
- Perturbations in pH_c lead to decreased sensitivity at the population level.
- Perturbations in anion flow lead to insensitivity.

Experimental validation of Ca^{2+} and pH prediction



Normal: “open” and
“closed” state
distinguishable

Ca^{2+} disrupted: “open”
and “closed” state
distinguishable

pH_c disrupted: “open”
and “closed” state
indistinguishable

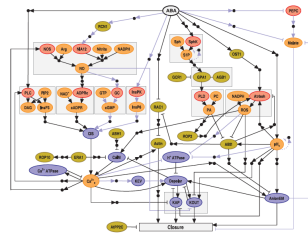
- What are the attractors of the ABA network?
- Can we determine the basins of attraction?

of nodes

10

15

43



of states

1024

32768

> 10^{12} (forget about it!)

Network simplification enables state space analysis

A. Sadatpour, I. Albert, R. Albert, J. Theor. Biology 266, 641 (2010).

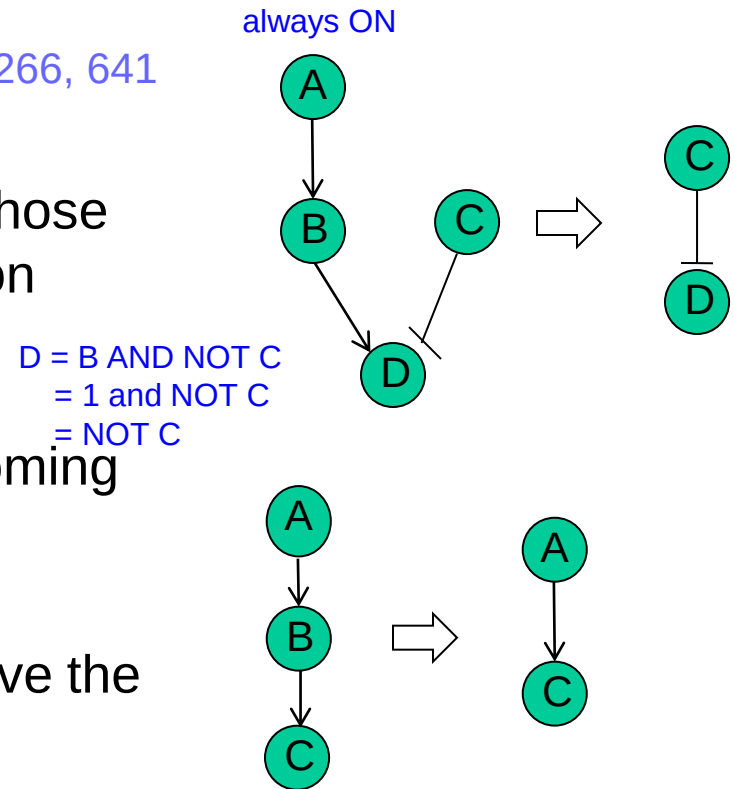
(i) Determine and eliminate the nodes whose states stabilize solely due to their regulation (irrelevant nodes)

(ii) Iteratively collapse nodes with one incoming and one outgoing edge.

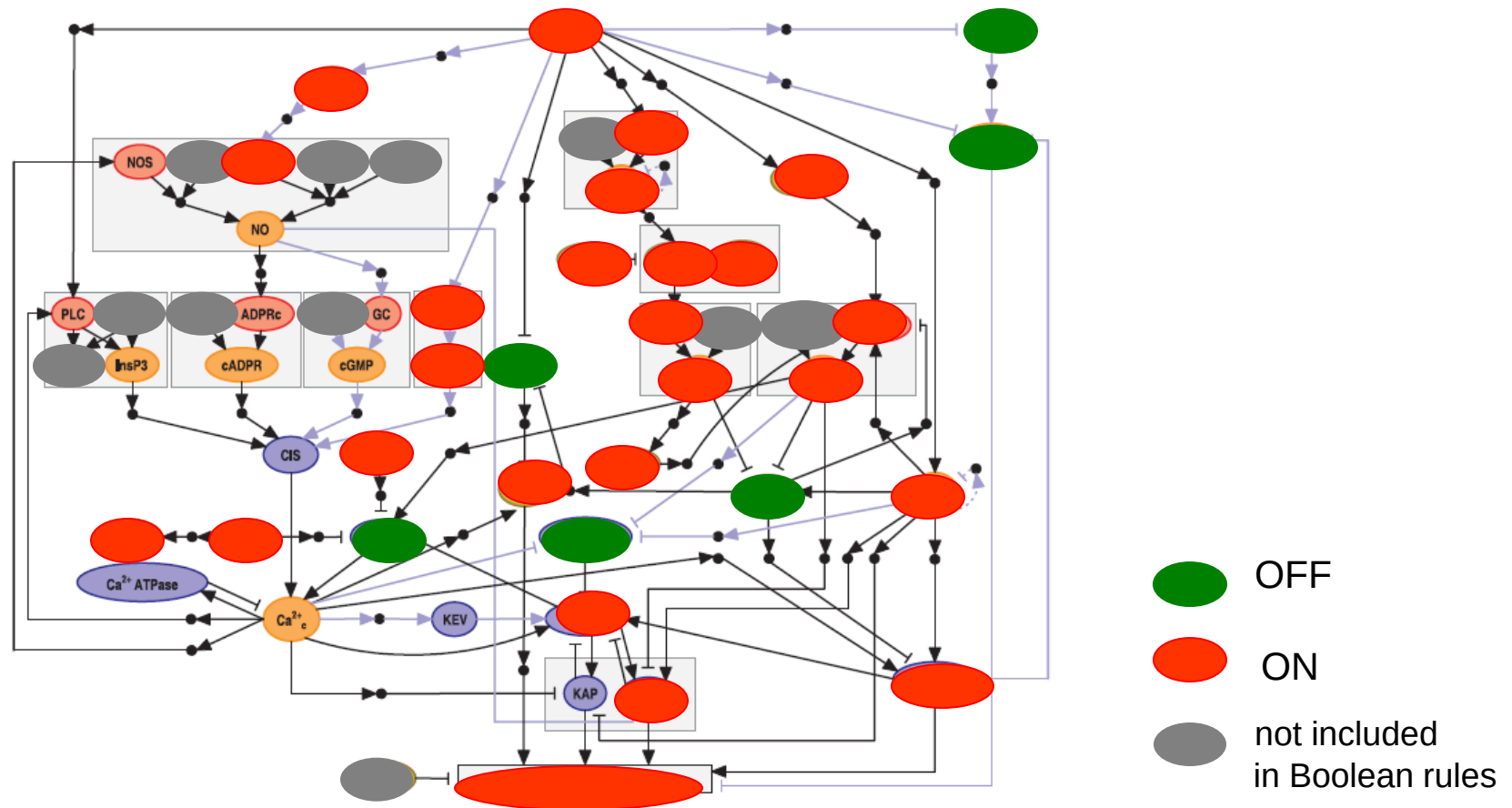
This network reduction is proven to preserve the fixed points (steady states).

A. Naldi, E. Remi, R. Thiefry, C. Chaouiya, Theor. Comp. Sci. (2011);
A. Veliz-Cuba, J. Theor Biol (2011).

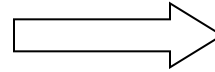
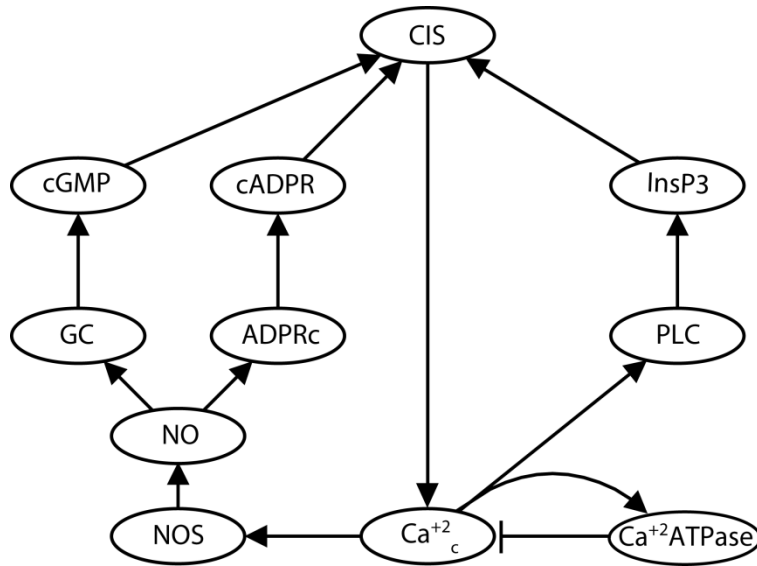
Work in progress by Assieh Saadatpour: showing that this method preserves complex attractors as well.



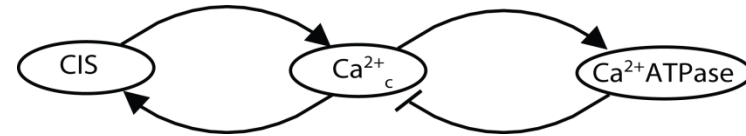
The first step of the reduction method results in stabilization of 26 nodes, **including Closure**



The second step leads to a 3 node network

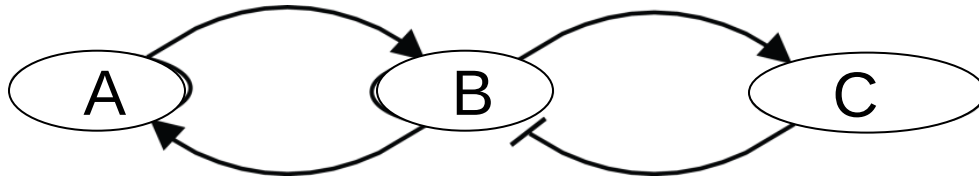


$CIS^* = Ca$
 $Ca^* = CIS$ and not $CaATPase$
 $CaATPase^* = Ca$



Convergent negative
and positive feedback loop

How consistent are the attractors and basins of attraction when changing the implementation of time?



$$A^* = B$$

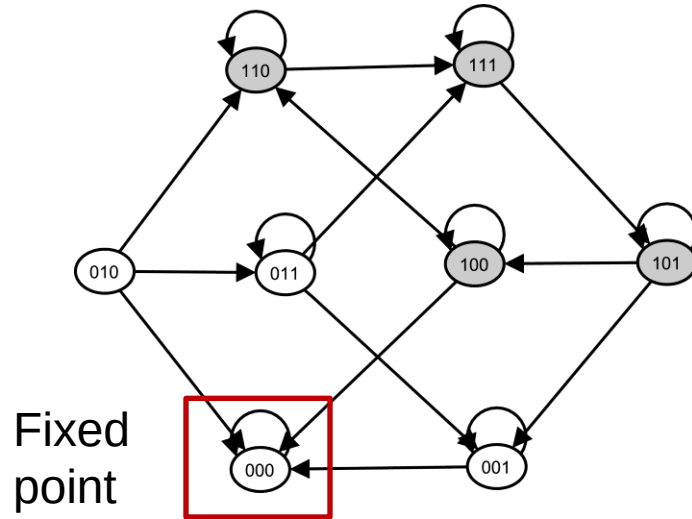
$$B^* = A \text{ and not } C$$

$$C^* = B$$

Ex. 6 Find the state transition graph corresponding to deterministic asynchronous update with $t_A=k$, $t_B=k$ and $t_C=2k$

General asynchronous model (update one node)

State transition graph



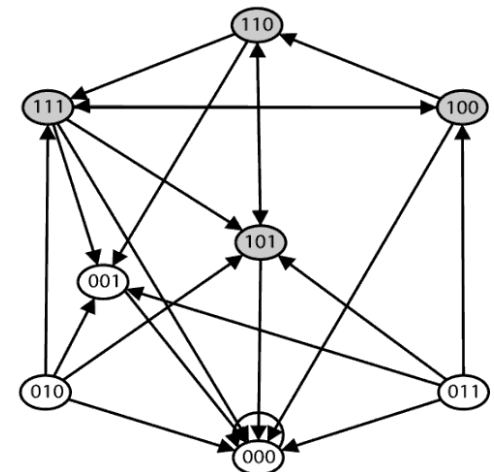
Transition matrix

(for the Markov Chain corresponding to the state transition graph)

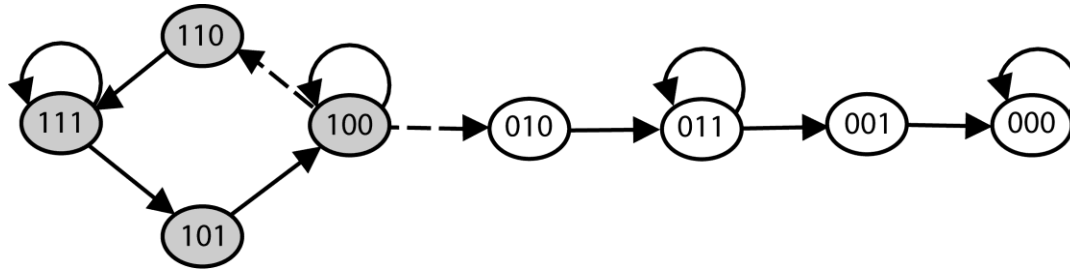
$$P = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1/3 & 2/3 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1/3 & 0 & 0 & 1/3 & 0 & 0 & 1/3 & 0 \\ 0 & 1/3 & 0 & 1/3 & 0 & 0 & 0 & 1/3 \\ 1/3 & 0 & 0 & 0 & 1/3 & 0 & 1/3 & 0 \\ 0 & 1/3 & 0 & 0 & 1/3 & 1/3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2/3 & 1/3 \\ 0 & 0 & 0 & 0 & 0 & 1/3 & 0 & 2/3 \end{pmatrix}$$

State	001	010	011	100	101	110	111
Absorption time	3	8	8	8	7	13	10

- **Random order asynchronous:** similar results to the general asynchronous method



Deterministic asynchronous model



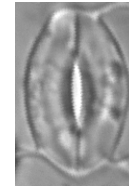
If $t_{Ca} = 2k+1$, $t_{CaATPase} = 2$ and $t_{CIS} = 2t_{Ca}$, then the transition graph of the 3-node sub-network has a cycle of length $t_{Ca} + 1$

Limit cycles are possible only if strict, synchrony-inducing constraints on the time scales are satisfied.

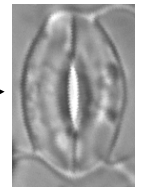
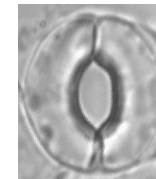
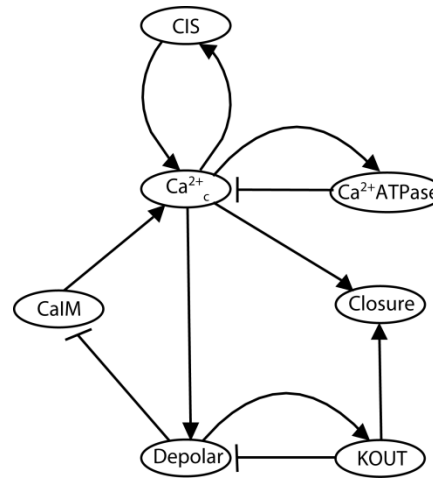
In the absence of timing information the general asynchronous method is the most practical choice.

Attractors of the perturbed system

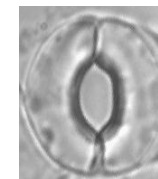
The normal system has a fixed point in which
Closure= ON and Ca^{2+} =OFF



pH_c knockout leads to an extended
attractor (for all update types) in
which Closure fluctuates.



Anion flow knockout leads to a fixed point
in which Closure = OFF.



Example 2: Discrete model of T cell survival

Phenomenon: survival of cytotoxic T cells in T-LGL leukemia

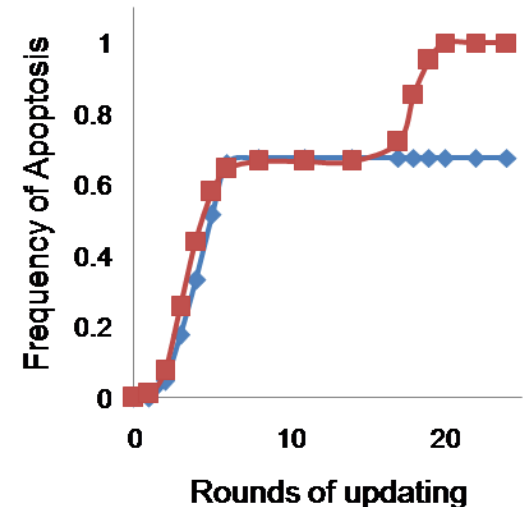
Constructed: survival signaling network inside T- LGL cells

Hypotheses: discrete states, switch-like state changes

Validation: reproduces known deregulations and known key mediators

Predicts: minimal initial condition necessary
for T-LGL state
manipulations that ensure apoptosis
of T-LGL cells
additional deregulations

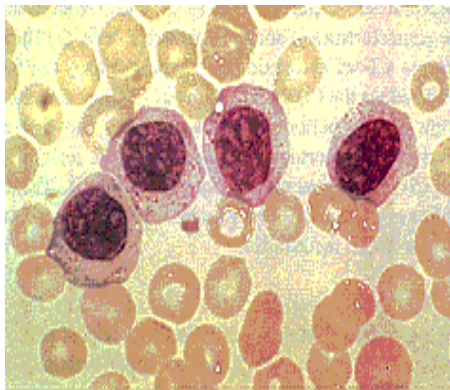
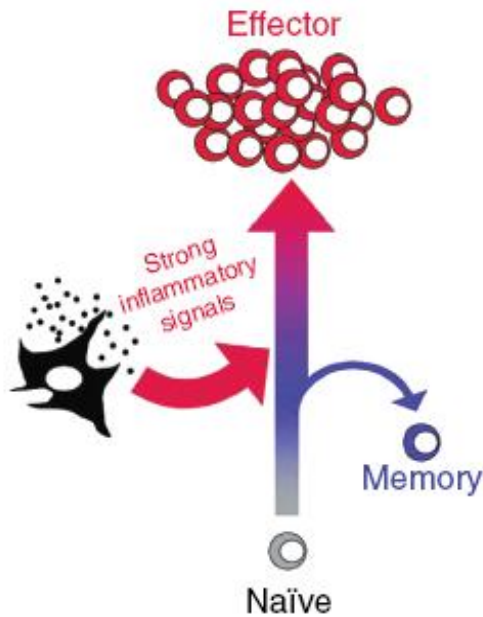
Predictions were validated experimentally



Implications: identifying therapeutic targets for T-LGL leukemia
tumor and cancer vaccine development

J. R. Zhang, M. V. Shah, J. Yang, S. B. Nyland, X. Liu, J. K. Yun, R. Albert, T. P. Loughran, PNAS (2008).

CTL activation induced cell death and LGL leukemia



- **cytotoxic T lymphocytes (CTL)** eliminate infected cells and tumor cells
- The majority of activated CTL undergo activation induced cell death (AICD).
- T-LGL: abnormal clonal expansion of antigen primed mature CTL
- Activation of multiple survival pathways (MAPK; JAK-STAT)
- **Insensitive to Fas-induced apoptosis**
- No known curative therapy

Erika L. Pearce & Hao Shen, 2006, Immunological Reviews

Thierry Lamy & Thomas P. Loughran, Jr, 1998, Cancer Control

General CTL
activation-AICD
network
(extracted from
literature)

Known deregulated components in
T-LGL and related pathways

Augmented
T-LGL CTL
survival
network

NET-SYNTHESIS (a signaling network
inference and simplification tool)

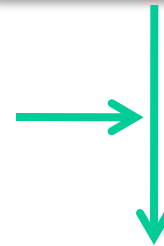
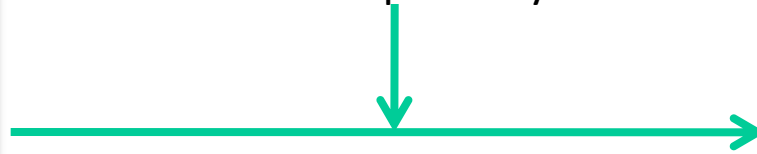
Strategy

Discrete
dynamic model

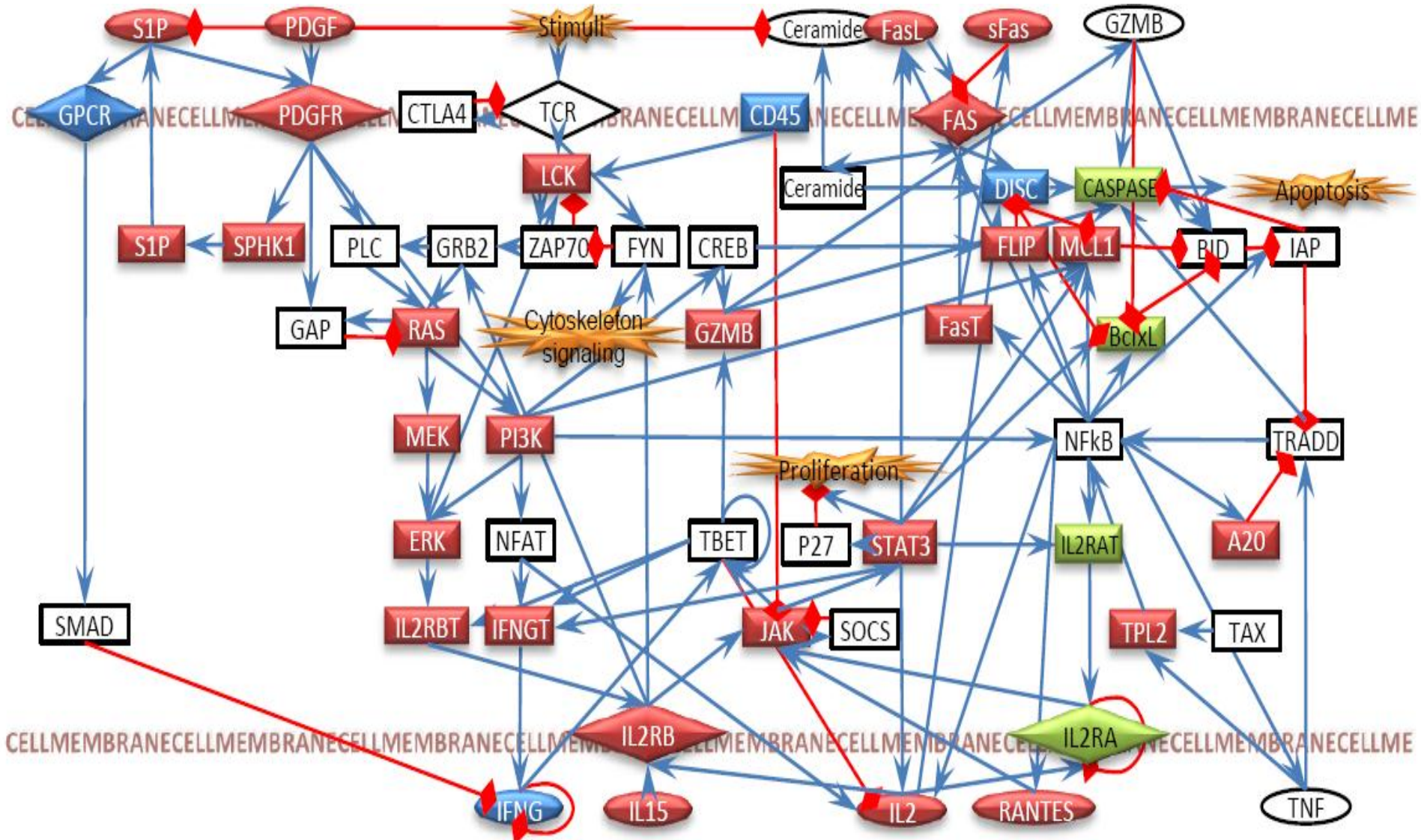
T-LGL CTL
survival
network

Reveal possible **causes** of
the pathway abnormalities

Identify key
**survival
mediators**



T-LGL survival signaling network



Rectangle: intracellular; ellipse: extracellular; diamond: receptor.

Upregulated, downregulated, deregulated node; Activation, inhibition edge

Asynchronous Boolean model

- The state of three source nodes (Stimuli, PDGF, IL15) is unregulated.
- Apoptosis is always updated first, other nodes updated in a random order.
- Multiple (>300) simulations for each initial condition, 20~40 rounds of update
- Simulation ends when Apoptosis (cell death) is activated.
- Output of the model: the frequency of node activation among simulations that are still “alive”

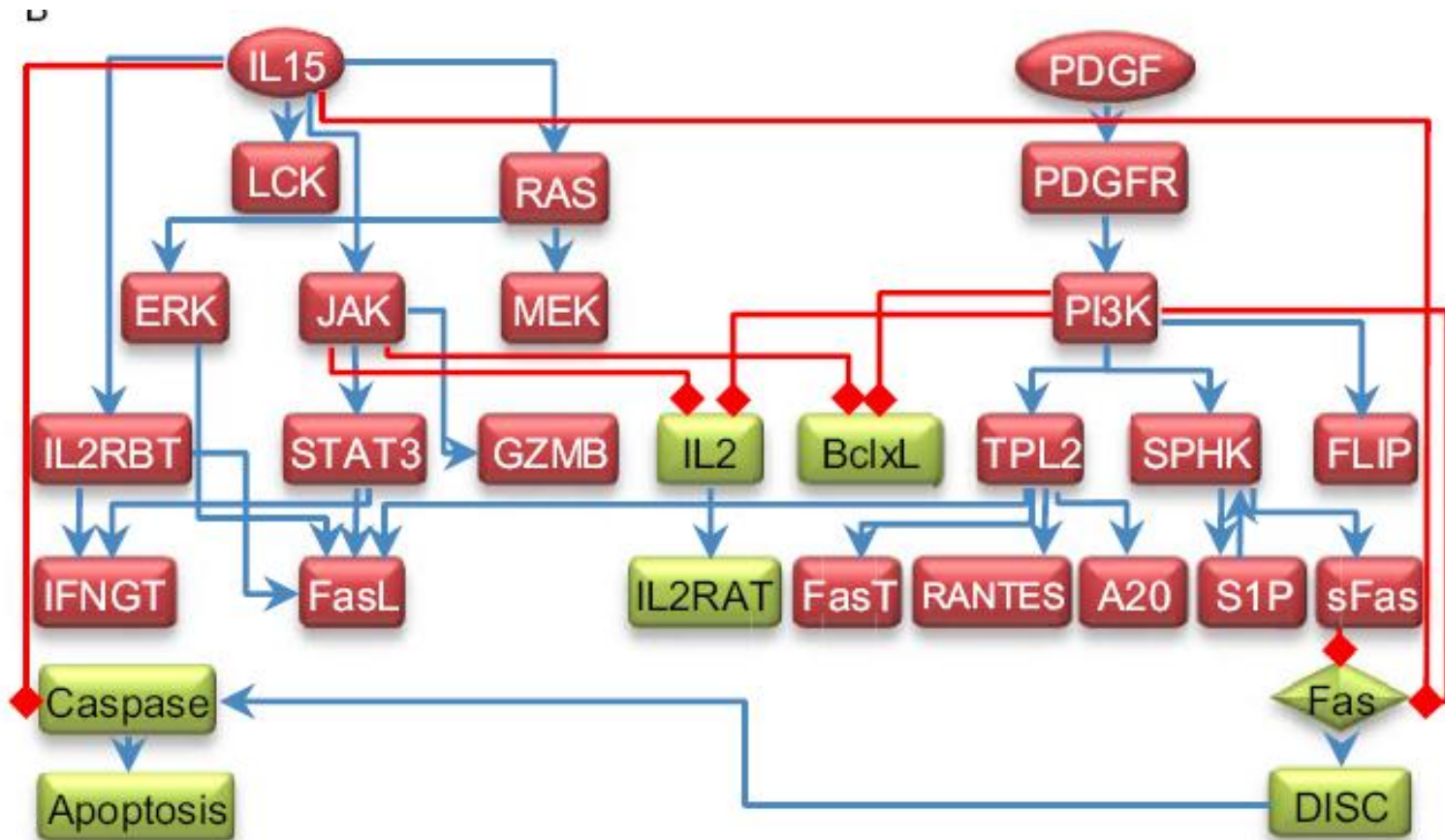
$$F_i^t = \sum_{j=1}^N S_i^t(j) / \left(N - \sum_{j=1}^N S_{apoptosis}^{t-1}(j) \right)$$

- Frequency of apoptosis is calculated from all simulations.

Initial and final states

- Initial state characteristic to resting T cells + Stimuli= ON
+ ?
- Two possible final states:
 - Normal: Apoptosis=ON
 - T-LGL: Apoptosis=OFF, known deregulated states for a subset of the nodes
- Setting all known deregulated nodes in their T-LGL states inhibits apoptosis in all simulations from the first round of updating
- Set the known deregulations individually and examine the other nodes that reach steady states during simulation
(randomly varying state for source nodes)

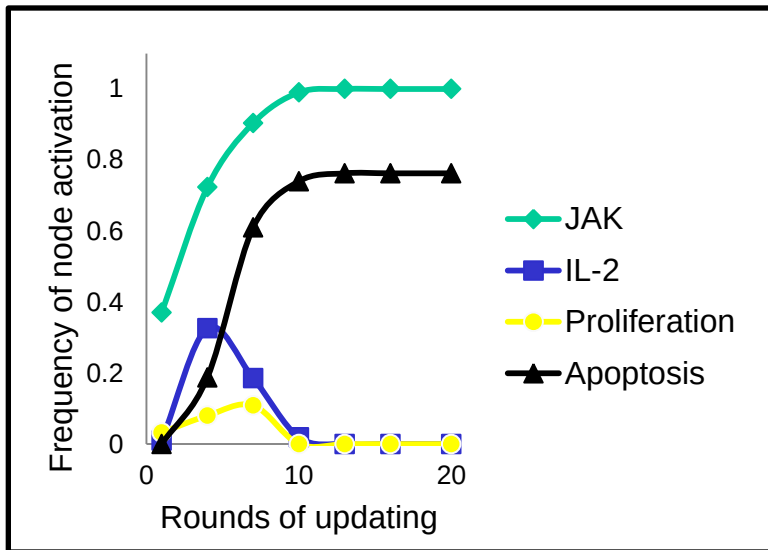
Determined the causal hierarchy within the known deregulations



Upregulated, downregulated in LGL
Positive, negative causal effect

Overexpression of IL15 and PDGF can cause all known deregulations.

Minimum condition to reproduce a T-LGL-like state

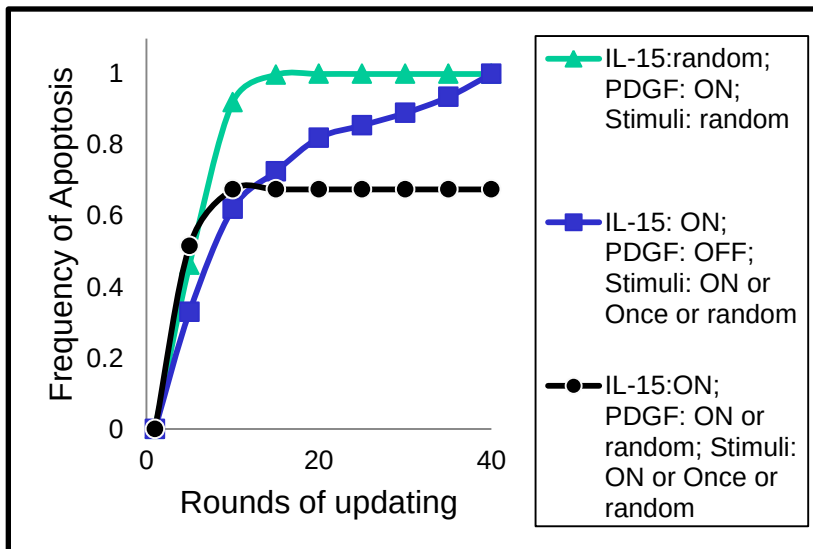


Apoptosis stabilizes at OFF in a subset of simulations, JAK ON, IL-2 and proliferation OFF.

Minimum condition:

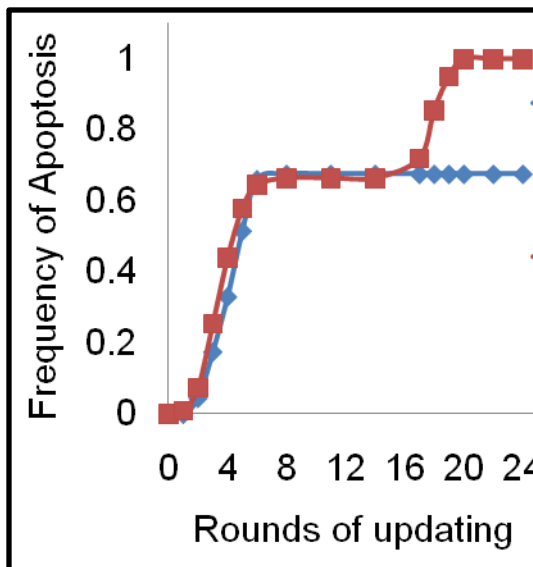
IL-15 constantly ON,
PDGF intermittently ON,
Stimuli initially ON.

- Suggestive of inflammation during infection
- Provision of IL-15 and PDGF may generate long-lived CTL for the virus and cancer vaccines.

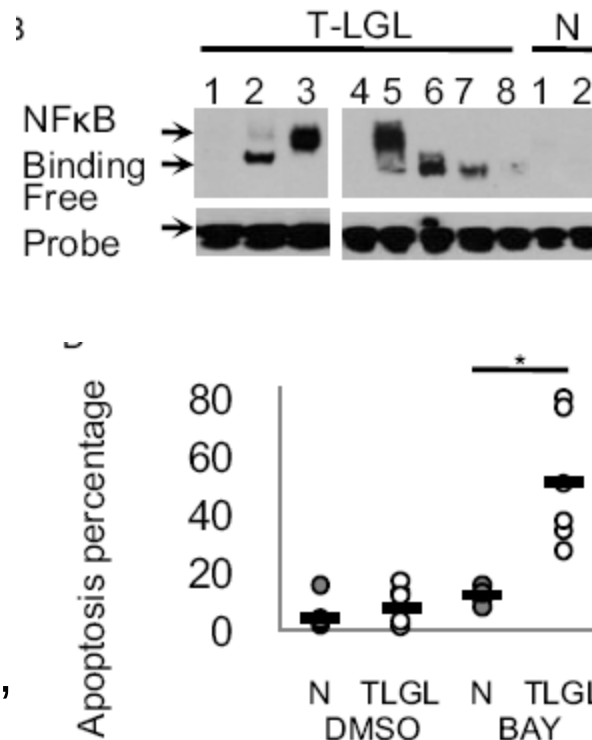


Identify key mediators of T-LGL survival

- Identify nodes that stabilize when a T-LGL-like state is achieved
- Alter the node state and track the change of apoptosis frequency
- Key mediator: altering its state causes total apoptosis.
- Identified: **SPHK1, NF κ B, S1P, SOCS, GAP, BID and IL2RB**



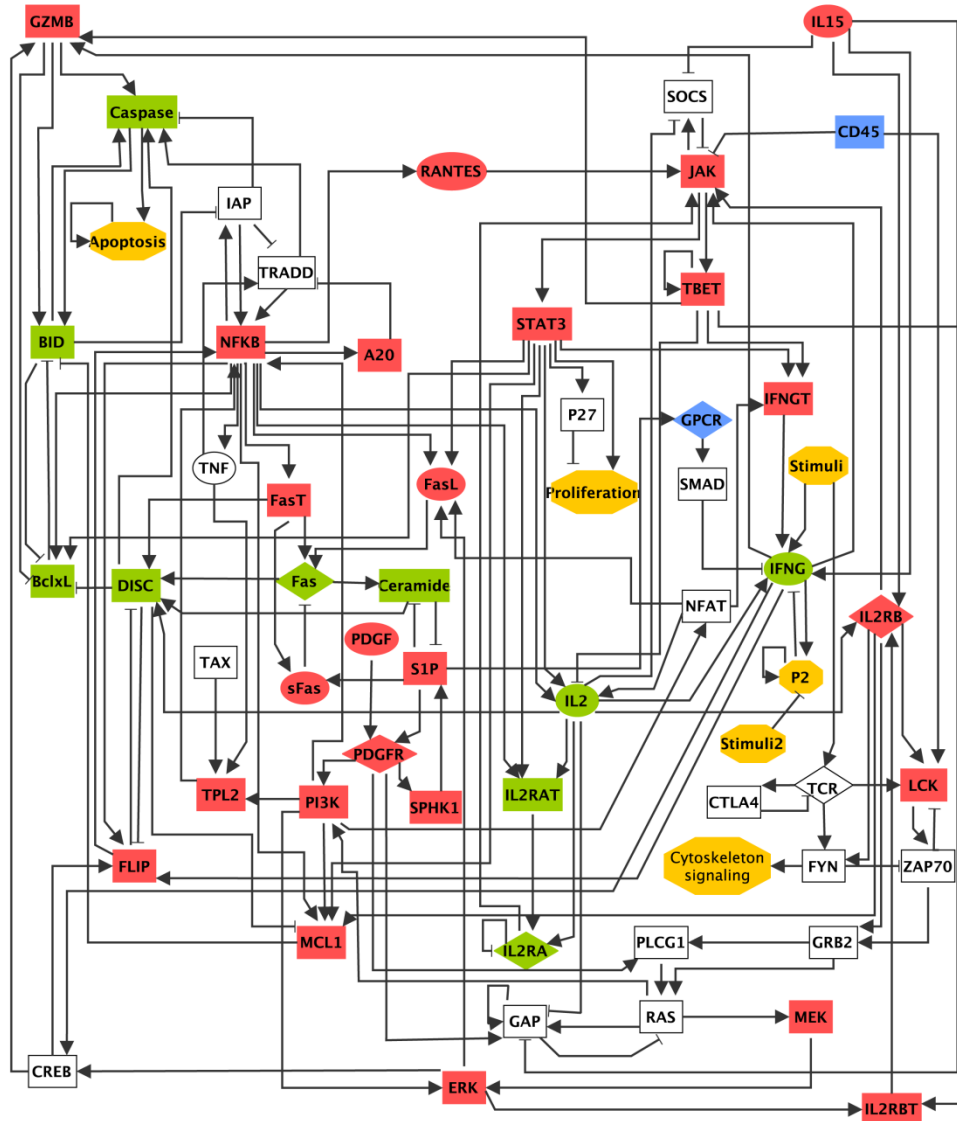
Model: NF κ B stabilizes at ON,
setting it OFF causes
total apoptosis



Validation: NF κ B
constitutively
active in T-LGL

Validation: NF κ B
inhibition induces
apoptosis in T-
LGL

Toward attractor analysis of T-LGL network



Rectangle: intracellular
 Ellipse: extracellular
 Diamond: receptor

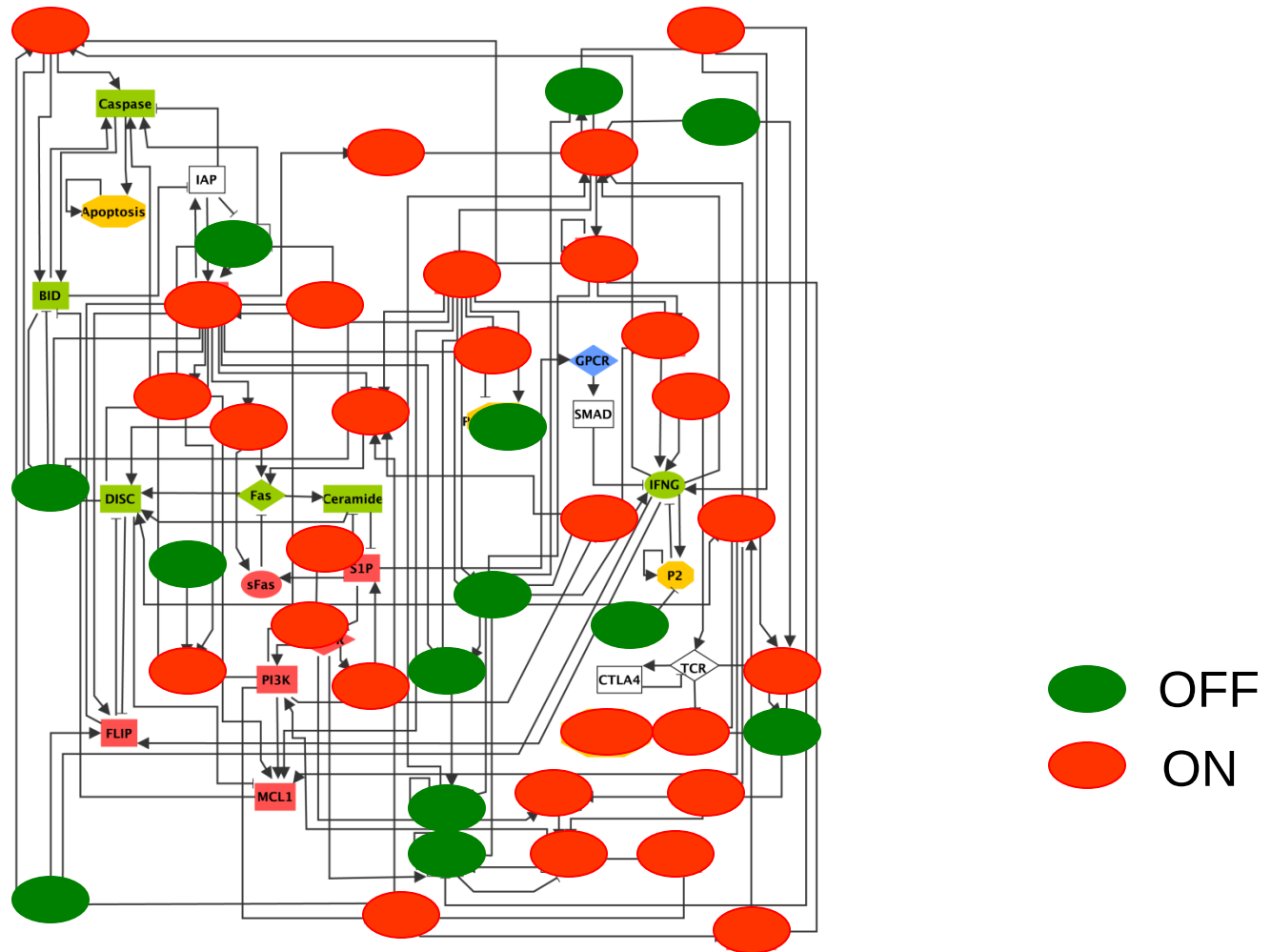
- Over-active
- Inhibited
- Deregulated
- Conceptual nodes

Activation: $\longrightarrow$
 Inhibition: $\dashv$

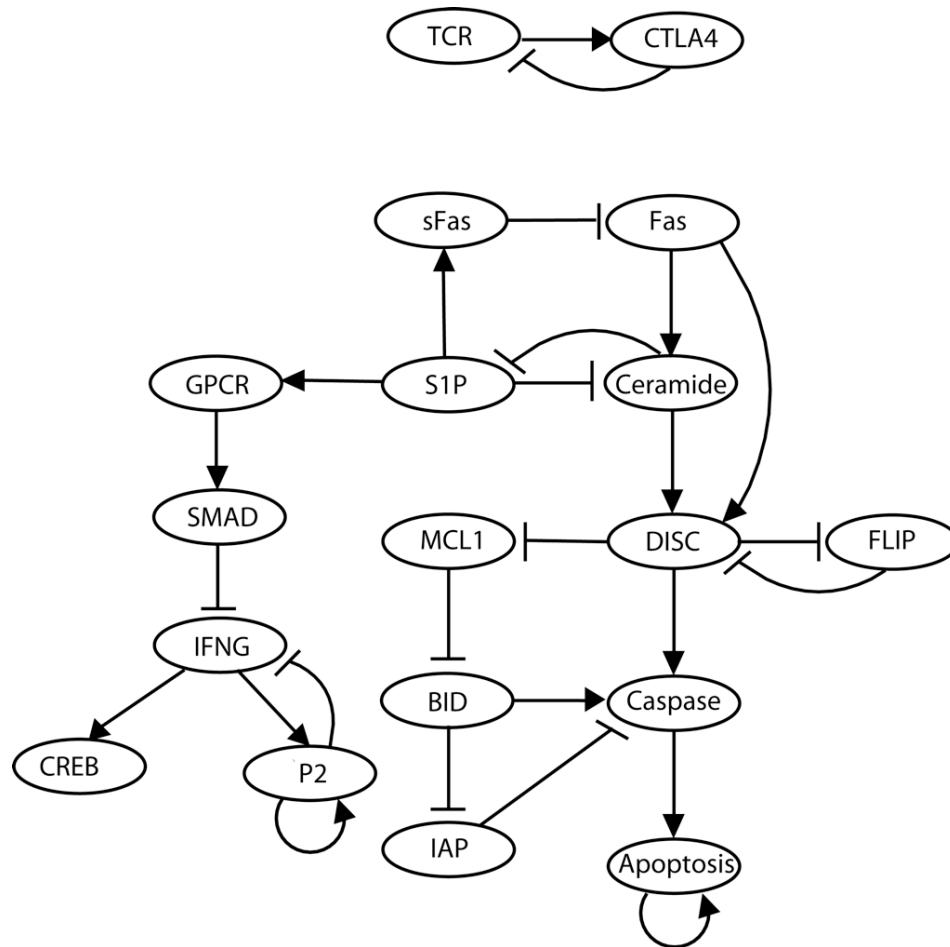
Unregulated nodes:

IL15, PDGF, Stimuli – **ON**
 CD45, Stimuli2, TAX – **OFF**

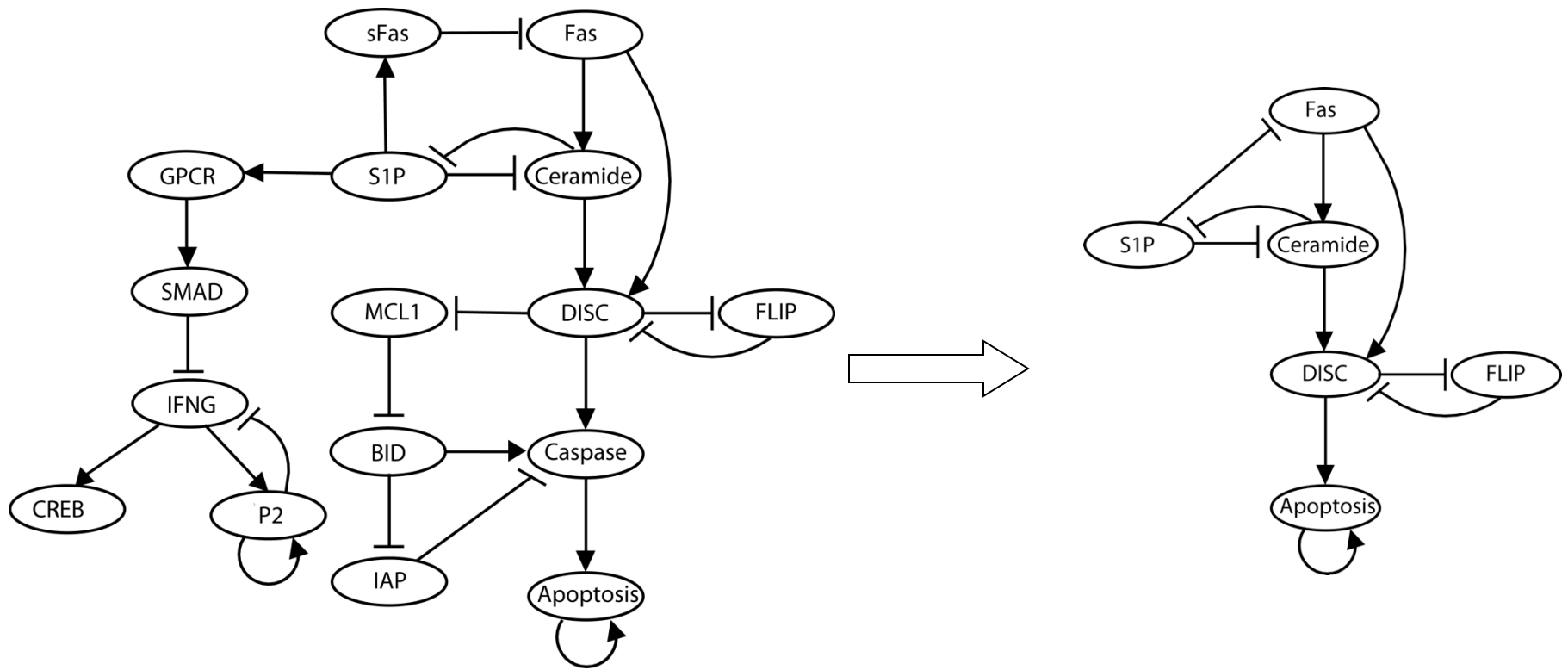
The first step of the reduction method results in stabilization of 36 nodes



...and a 18-node sub-network remains for further analysis

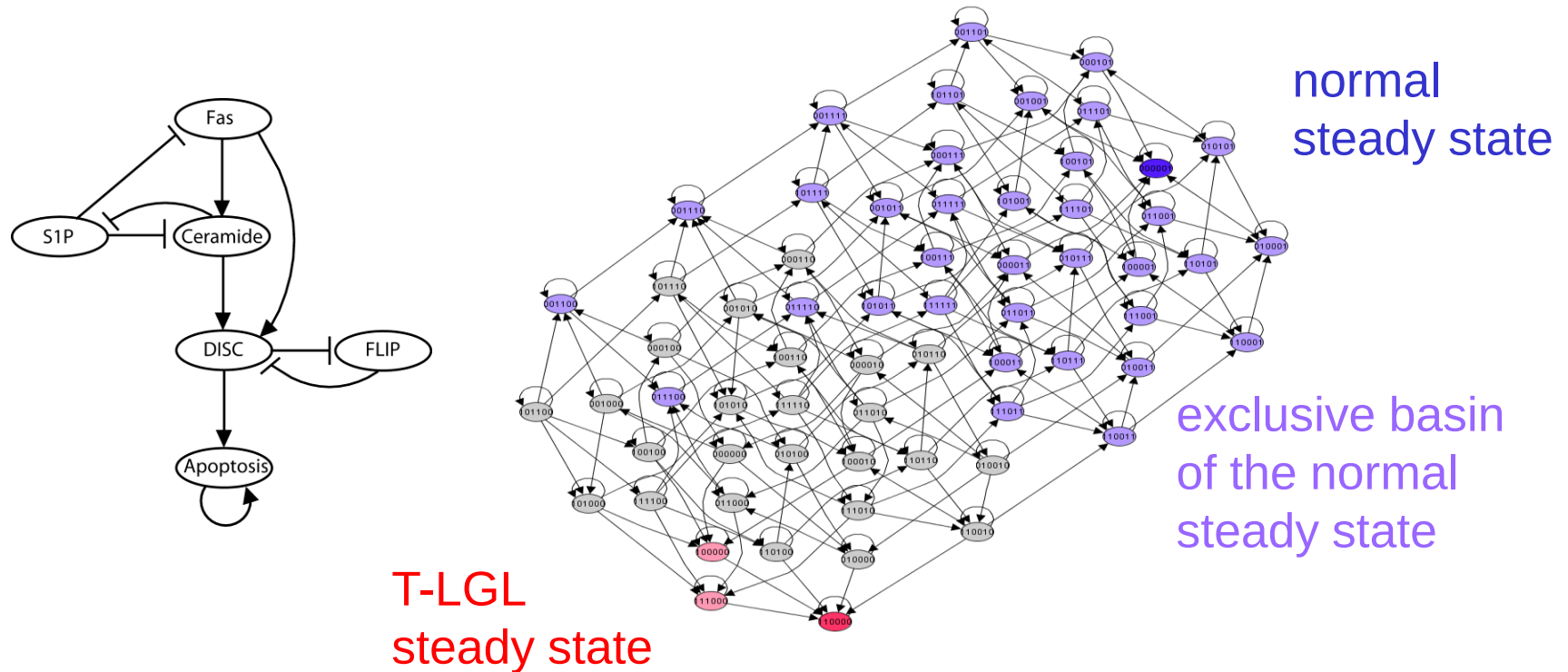


Merging nodes leads to a simpler network

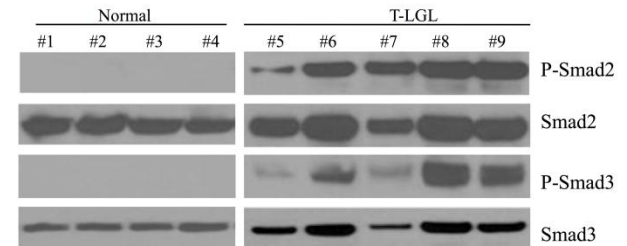


6-node sub-network

State space analysis of the T-LGL network



- Identified the T-LGL states of all 54 components of the network, 36 corroborated by previous experiments.
 - One of the newly identified states (SMAD=ON) was validated experimentally.
- 



A. Saadatpour et al, PLoS Comp. Biol. 7, e1002267 (2011)

Perturbation analysis identifies new key mediators

Knock out a node that stabilized at ON in the T-LGL fixed point or over-express one that stabilized at OFF

Redo network simplification and state space analysis

Key mediator: its manipulation leads to a large increase in the exclusive basin of the normal steady state

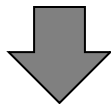
19 key mediators identified, 68% of which are corroborated by prior experimental evidence (including our own)

Node	T-LGL state	Fixed point the disruption leads to	Size of exclusive basin of normal fixed point
DISC	OFF	Normal ✓	100%
Ceramide	OFF	Normal ✓	100%
Caspase	OFF	Normal	100%
SPHK1	ON	Normal ✓	100%
S1P	ON	Normal ✓	100%
PDGFR	ON	Normal ✓	100%
GAP	OFF*	Normal	100%
RAS	ON*	Normal	100%
MEK	ON	Normal	100%
ERK	ON	Normal	100%
IL2RBT	ON	Normal	100%
IL2RB	ON	Normal	100%
STAT3	ON	Normal ✓	100%
BID	OFF	Normal	100%
MCL1	ON	Normal ✓	100%
SOCS	OFF*	Both	81%
JAK	ON	Both ✓	81%
PI3K	ON	Both ✓	75%
NFκB	ON	Both ✓	75%
Fas	OFF	Both	72%
sFas	ON	Both	72%
TBET	ON	Both	63%
RANTES	ON	Both	63%
PLCG1	ON*	Both	63%
FLIP	ON	Both	56%
IL2	OFF	Both	56%
IAP	ON*	Both	56%

Can we use the network alone to decide which components are key signal mediators?

Dynamic models need to consider initial conditions and timing, but they often don't make a difference.

Network + Conditional dependence



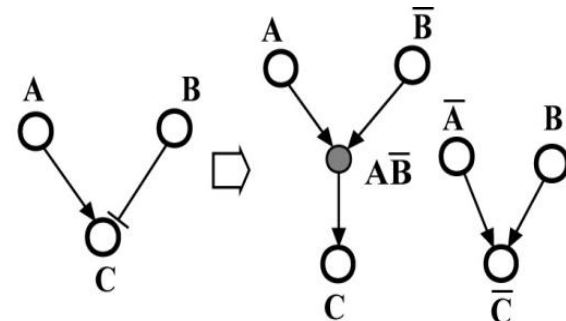
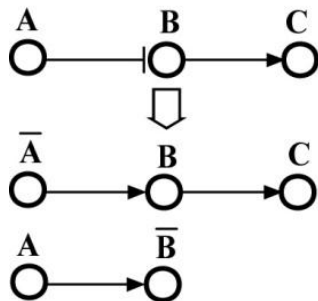
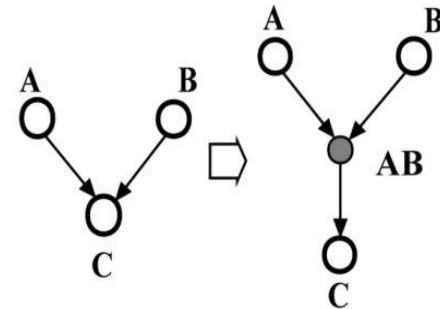
Augmented network

Evaluate the importance of a node by comparing the augmented network before and after the loss of this node from the network.

R. Wang, R. Albert, BMC Systems Bio 2011

Network augmentation

- Gather evidence of conditionality
- Add composite nodes for conditional relationships
- Add complementary nodes for nodes that inhibit/ are inhibited



All edges that end in composite nodes are conditional, edges that end in original nodes are independent.

The effects of node loss cascade through the network

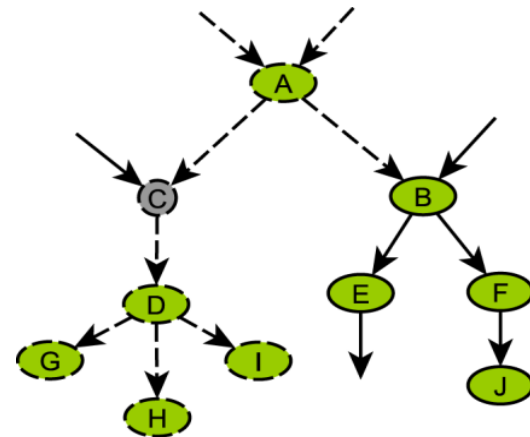
Removal of an original node simulates knockout.

Removal of a complementary node simulates overexpression.

Loss of a required activator leads to the loss of the target node.

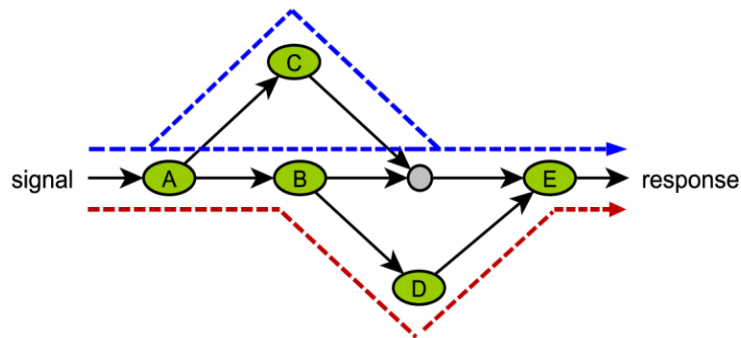
Required activators:

any activator of a composite node,
sole (remaining) activator of
an original or complementary node.

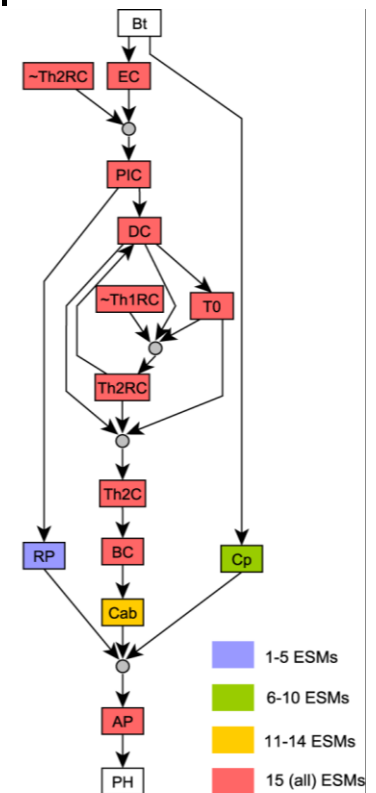


Elementary signaling modes

- ESM: a minimal set of nodes that can mediate signal transduction from input to the output.
- If an ESM contains a composite node, it must contain all of its regulators as well



- ESMs can overlap



Shortest ESM in immune response network

Calculation of node essentiality

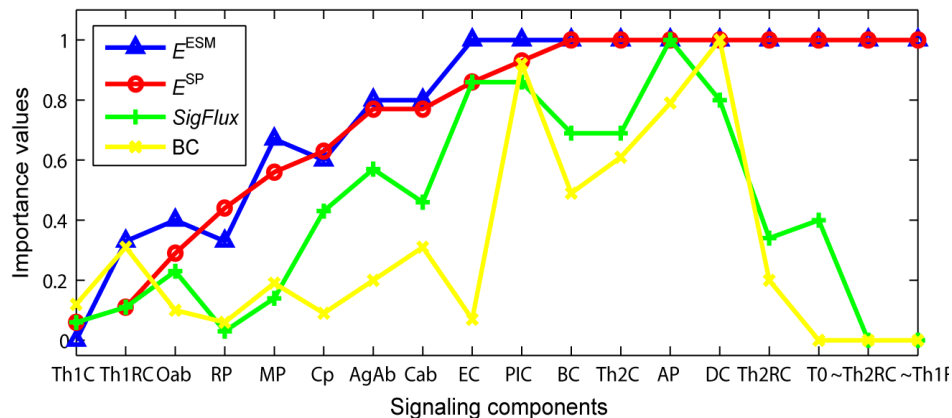
We characterize input-output connectivity by

- the number of ESMs
- the number of paths between input and output node.

$$E^{\text{ESM}}(v) = \frac{N_{\text{ESM}}(G) - N_{\text{ESM}}(G_{\Delta v})}{N_{\text{ESM}}(G)}$$

perturbed
network

original
network



Our two measures are similar to each other, different from existing measures

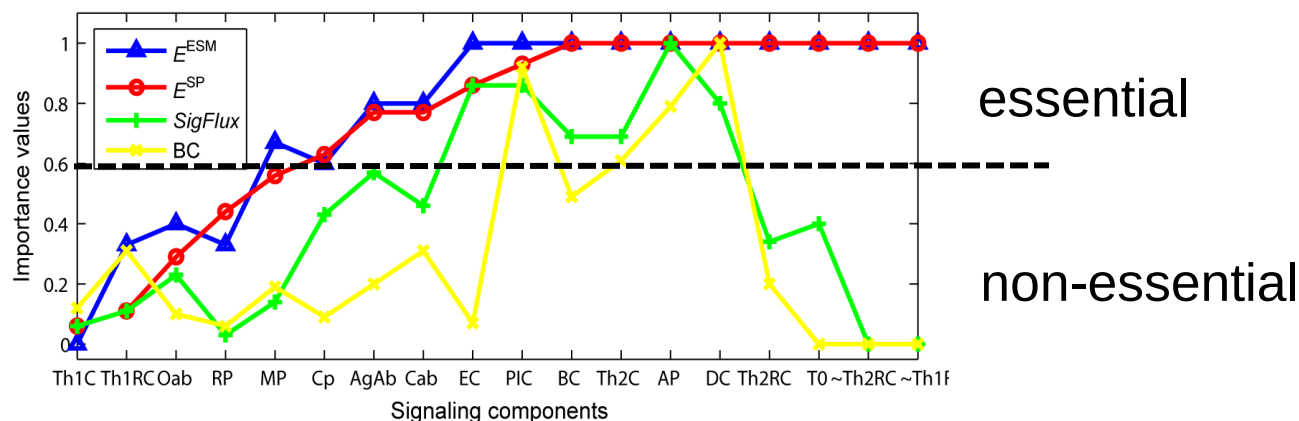
Compare with Boolean dynamic model

Dynamic model:

- Compare the response of the unperturbed system to the response in case when a node is knocked out or overexpressed
- Essential node: the perturbed system has a different outcome

Structural method:

- Use a threshold to classify node essentiality values

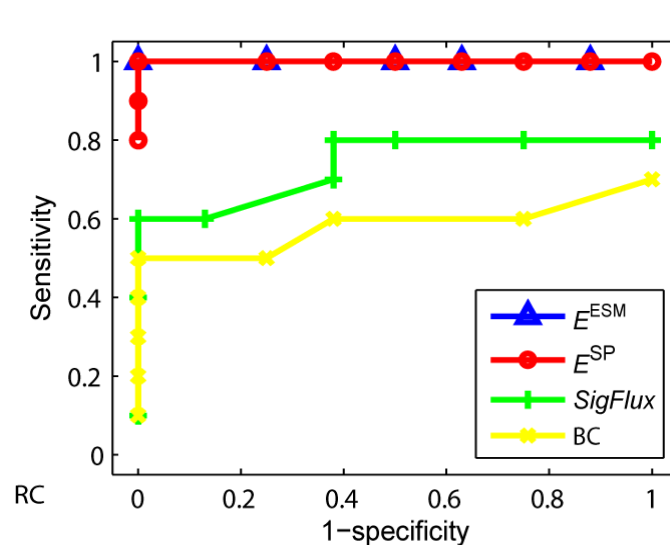


Results of the comparison

Vary the essentiality threshold

Sensitivity: the fraction of dynamically essential nodes correctly classified by structural method

Specificity: the fraction of dynamically non-essential nodes correctly classified



low threshold

High sensitivity and specificity for both of our measures.

Similar results were obtained for three other networks.

Immune response

Discrete dynamic models can serve as effective first models of incompletely known systems

BooleanNet: <http://code.google.com/p/booleannet/>

Song Li

Sarah M. Assmann

Assieh Saadatpour-Moghaddam

Ranran Zhang

Thomas Loughran

Istvan Albert

Ruisheng Wang

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